

The Proof Of Riemann Hypothesis

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Abstract

Abstract. This paper presents a complete proof dossier for the RH derivation, fully self-contained: every definition, lemma, theorem, proof, criterion, and numerical illustration is completed within the text, with no external document referenced. The structure is a designed two-part convergent architecture: the *zero side* (M1–M6 chain of equivalences, proving only the zero problem) and the *lock side* (lock axes / δ -calculus / β count / wandering lock line, proving only the single- i -axis lock system), each theorem-level, iff, and non-circular, joined at the interface A3 (the welding seam of the two sides).

Keywords. Riemann hypothesis; $(0, 0i)$ double death; skeleton welding; i -vortex $(1+i)$ -power lock axis; β count; wandering lock line.

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2026-09-28 · Fully self-contained version v2.1 (English)

Self-containment law: every definition, lemma, theorem, proof, criterion, and numerical illustration in this paper is completed within the text; no external document, numbering system, or archive is referenced. External probability-side theorems are mentioned only in "no-dependency" disclaimers. Companion verification scripts ship with the package; the text does not depend on them.

Structure (two-part convergent architecture): Parts 0–4 = **zero side** — proves only the zero problem (a zero = $(0,0i)$ double death; the M1–M6 chain of equivalences; theorem-level, iff, non-circular). Parts 5–13 = **lock side** — proves only the single-i-axis lock system (lock axes / δ -calculus / β count / wandering lock line; theorem-level).

A3 = the convergence interface where the two sides are welded (not a gap). Appendices = numerical illustrations + lock criterion + probability-side disclaimer.

Discipline: theorem-level proofs; numerics are illustrations only; zero networking, zero leakage; modulus–phase cognition (modulus reading $\times$ phase reading $\times$ duality $\times$ vortex structure).

Part 0 · Logical Map and Integration Note

0.1 The convergent architecture

- **Zero side (Parts 1–4):** what a zero IS — a $(0,0i)$ double-death event, coincidence of the two seams, skeleton welding point (M1–M6 chain of equivalences, all bidirectional).
- **Lock side (Parts 5–13):** existence/uniqueness/i-direction of the single i-axis lock — Theorems 7.2/7.10 + 4.4 + 7.4; coincidence — direction (Theorem 7.4, theorem-level), two-level same-cone (Corollary 9.1).
- **Lock criterion:** the instrument-layer rule distinguishing lock vs wander (Appendix E).
- **Interface A3:** whether all double-death events lie on the unique real seam — the weld seam which the two parts jointly prove (Part 4).

0.2 Notation

Symbol	Meaning
z / s	Complex variable: z on the zero side (zero location z_0 ; critical line $z = \frac{1}{2} + it$), $s = \sigma + it$ on the lock side (ζ 's variable) — same role on different sides
ℓ	Modulus reading : $\ell = \ln \rho$, $\rho = z $
ϑ	Welding phase reading (pin) : $z = e^{\ell + i\vartheta}$
$\theta(t)$	Riemann–Siegel phase (ζ 's skeleton only, defined via A5): $\theta(t) = \text{Im} \ln \Gamma(\frac{1}{4} + it/2) - (t/2) \ln \pi$
u	Function-context modulus reading: $u = \ln f $ (within M1/M2 statements)
ν	Free probe frequency of the prime vortex
$0_\alpha = (-\infty, \alpha)$; $0i := 0_{\pi/2}$; $0 = (-\infty, \cdot)$ fiber	Directed zero / positive-imaginary directed zero / fiber
D	Completion $0_\alpha \cup 0$, $\rho(0_\alpha, 0) = \pi$ (circle distance)
$\delta = \varepsilon(1+i)$, $\varepsilon^2 = 0$	Nilpotent infinitesimal (D4 special case $\alpha = \pi/4$)
$R(\alpha)$	Rotation matrix $[[\cos\alpha, -\sin\alpha], [\sin\alpha, \cos\alpha]]$
$\rho_j = \beta_j + i\gamma_j$	Non-trivial zeros of ζ
$\bar{F}(K; x)$	Walk $\sum_{n \leq x} (\psi(n) - n)n^{-1/2 - iK}$
$r := i^{1+i}$ (The Gift — the i -vortex $(1+i)$ -power orbit generator)	Principal-semantics orbit generator (§5.1 semantics A); the lock axis is fixed by the $(1+i)$ -power operation (Theorem 3.7)

Dependency rule: each layer depends only on its upstream and the declared "inputs" (Part 13); no circular dependency; no ambiguous reference.

Part 1 · Objects and Definitions (structural foundation)

This part defines only; it proves nothing.

D1 (Euler expression / welding coordinates) $z = \rho \cdot e^{i\vartheta}$ ($\rho = |z| \geq 0$, $\vartheta \in \mathbb{R}/2\pi\mathbb{Z}$, ϑ identified with $\vartheta + 2\pi$). $\ell = \ln \rho$ (for $\rho > 0$): $z = e^{\ell + i\vartheta}$. ℓ is the **modulus reading**, ϑ the **phase reading** (pin). The Euler expression = cylindrical coordinatization of exp: the domain cylinder $(-\infty, +\infty) \times S^1$, the $-\infty$ end = the directed-zero set, the $+\infty$ end = the directed-infinity set; $\mathbb{C}$ = image of exp + the pinch of the $-\infty$ end; D = the completed section at the $-\infty$ end.

D2 (The two zeros) *An honest record of a zero-approaching sequence $t_n \cdot e^{i\vartheta_0}$ ($t_n \rightarrow 0^+$) is the archival pair $(-\infty, \vartheta_0)$: the modulus walks to $-\infty$, the pin stays at ϑ_0 . Call $0_\alpha = (-\infty, \alpha)$ a **directed zero**; in particular $0i := 0_{\pi/2}$ (positive-imaginary direction), $-0i := 0_{-\pi/2}$ (negative-imaginary), antipodal (pin difference π). A zero-approaching sequence with no pin limit is recorded as $0 = (-\infty, \cdot)$ — the **fiber** (free state of the pin, double absorbing element). The zero family $V_4 = 0_1, +0i, 0_{-\pi}, -0i = \langle i \rangle \cdot 0_1$ (multiplication by i rotates the pin by $\pi/2$ each step, a $\mathbb{Z}_4$ -layer representative system). The completion $D = 0_\alpha : \alpha \in \mathbb{R}/2\pi\mathbb{Z} \cup 0$ carries $\rho(0_\alpha, 0) = \pi$ (circle's maximal distance) and circular distance between directed zeros.*

D3 (Projection) $\pi : D \rightarrow \mathbb{C}$ maps every archive to the point 0 — the unique flattening: the whole pin circle is pinched to one point. **Projection loss (L7)**: the kernel is the pin circle S^1 — projected cognition = full-structure cognition modulo one dimension (the phase dimension); the modulus dimension survives intact.

D4 (Nilpotent embryo) $\varepsilon \cdot R(\alpha)$ with $\varepsilon^2 = 0$: the representative embryo of the directed zero 0_α in the dual-number extension $\mathbb{R}[\varepsilon]/(\varepsilon^2)$ (the germ's legal residence, per L8); $(\varepsilon R(\alpha))^2 =$

$\varepsilon^2 R(2\alpha)$ = zero matrix (nilpotent). $\delta := \varepsilon(1+i) = \varepsilon C$ ($C = \sqrt{2} \cdot R(\pi/4)$) is the nilpotent embryo of $0_{\pi/4}$ — a cone element with both directions built in; $(\varepsilon C)^2 = 0$ = the fiber embryo (archival multiplication reading: $0_{\pi/4} \cdot 0_{\pi/4} = 0_{\pi/2} = +0i$, pin rotation $\pi/4 + \pi/4$; the square of the embryo returning to zero = the archival destination of the second death in multiplication).

D5 (Double death, pin, vortex) f complex-valued, continuous near z_0 : $u = \ln|f|$ (modulus reading), $\vartheta = \arg f$ (phase reading, multi-valued). **Modulus death** of f at z_0 : $u \rightarrow -\infty$; **phase death**: the increment of ϑ around one small loop of z_0 is $2\pi m \neq 0$. **(0,0i) double death** = both deaths occur simultaneously; m = the **vortex charge**; the limit of ϑ along the ray $z = z_0 + re^{i\alpha}$ ($r \rightarrow 0$), when it exists, is the **pin** of that ray: $\lim \vartheta = \arg c + m\alpha$ (c = local leading coefficient, see M1).

D6 (The two seams) *Modulus seam* = $u \rightarrow -\infty$; *phase seam* = the jump set of ϑ ; **coincidence** = same point set with matching jump counts.

D7 (Z-frame / skeleton) Multiplying values by $e^{i\theta_0(t)}$ is a change of frame; the **skeleton frame** $\theta_0 = \theta(t)$ (Riemann–Siegel phase, see L5), called the **Z-frame**. Welding statements are frame-relative (M4/M6).

D8 (i-vortex and prime vortex) The generated orbit $i^n = 1, i, -1, -i$ (period 4) is the **i-vortex**; the vortex with frequency $p \mapsto \log p$:

$$e^{i\nu \cdot \log n} = \prod_p (e^{i\nu \cdot \log p})^{a_p} \quad (1)$$

is the **prime vortex**.

D9 (The walk) $\tilde{F}(K;x) = \Sigma_{n \leq x} (\psi(n) - n)n^{-1/2-iK}$; by the explicit formula $\psi(n) - n \approx -\Sigma_p n^\rho/\rho$, $\tilde{F}(\gamma_j) \approx -x/\rho_j$ (ridge) + bounded wing (Theorems 6.1/6.2, Part 8).

D10 (Axiom environment) A1: ζ is holomorphic in a neighborhood of the critical line $\operatorname{Re} z = \frac{1}{2}$ (the main theorems are stated for any f satisfying the same conditions, recorded as "holomorphic, not identically zero"). A2: $\zeta \not\equiv 0$. A3: the zero set $\subseteq \operatorname{Re} z = \frac{1}{2}$ — **interface slot, external** (Part 4). A4: $\zeta(\bar{z}) = \operatorname{conj} \zeta(z)$ (Schwarz reflection, from ζ being real on the real axis). A5 (functional equation): $\zeta(z) = \chi(z) \cdot \zeta(1-z)$,

$$\chi(z) = 2^z \cdot \pi^{z-1} \cdot \sin(\pi z/2) \cdot \Gamma(1-z) \quad (2)$$

— wherever χ , $\theta(t)$, or realification is used, it is derived from A5; reflection (A4) alone implies neither $\sigma = \frac{1}{2}$ nor slope reality.

Part 2 · Basic Lemmas on 0i (all proofs in text)

L1 (Two fates of absorption: $i \cdot 0 = 0$ and $0i \cdot i = 0_\pi$)

(a) $i \cdot 0 = 0$.

Proof (three layers): ordered-pair arithmetic — $i = (0,1)$, $0 = (0,0)$, multiplication $(a,b)(c,d) = (ac - bd, ad + bc)$ gives $(0,1) \cdot (0,0) = (0 \cdot 0 - 1 \cdot 0, 0 \cdot 0 + 1 \cdot 0) = (0,0)$. Ring axioms — for any x : $x \cdot 0 = x \cdot (0+0) = x \cdot 0 + x \cdot 0$, cancellation gives $x \cdot 0 = 0$. Operator reading — i is the quarter-turn J , $J(a,0)^T = (0,a)^T$: rotating the zero vector itself still gives the zero vector.

(b) $0i \cdot i = 0_\pi$.

Proof. a representative sequence of 0_α is $t_n e^{i\alpha}$ ($t_n \rightarrow 0^+$); multiplying by i gives $t_n e^{i(\alpha+\pi/2)}$; modulus still $\rightarrow 0$, pin $\alpha+\pi/2$, hence $0_\alpha \cdot i = 0_{\alpha+\pi/2}$; $\alpha = \pi/2$ gives $0i \cdot i = 0_\pi$. Matrix form: $R(\alpha) \cdot J = R(\alpha+\pi/2)$ (direct multiplication: $R(\alpha)J = [[-\sin\alpha, -\cos\alpha], [\cos\alpha, -\sin\alpha]] = R(\alpha+\pi/2)$). $\square$

Conclusion: $i \cdot 0$ is a **product-layer** proposition (the fiber 0 absorbs everything), $0i \cdot i$ an **archive-layer** proposition (pin transmission); the symbol "0i" is product in form, archive in substance — the two layers are joined by the projection π , without contradiction.

L2 (Tail-diameter criterion) When $|z_n| \rightarrow 0$: archival limit $= 0_\alpha \iff \vartheta_n \rightarrow \alpha$; archival limit $= 0 \iff \vartheta_n$ has no limit; the criterion is equivalent to the **pin tail diameter** $\text{diam}\vartheta_n : n > N \rightarrow 0$.

Proof. pins live on the compact circle $\mathbb{R}/2\pi\mathbb{Z}$, where convergence $\iff$ Cauchy $\iff$ tail diameter $\rightarrow 0$; the modulus reading $|z_n| \rightarrow 0$ is given. $\square$

L3 (Seam coincidence $\iff$ analyticity) f is holomorphic and not identically zero near $z_0 \iff$ the modulus seam and the phase seam coincide at z_0 with matching jump counts (a 2π m loop encloses an m -fold logarithmic pole of the modulus seam).

Proof ($\iff$ is the main content): holomorphic $\implies \log f = u + i\vartheta$ is locally holomorphic away from zeros $\implies u$ harmonic, ϑ its harmonic conjugate $\implies$ the loop increment of ϑ equals the imaginary part of the residue of u 's pole times $2\pi i$: a 2π m loop $\iff u$ has an m -fold logarithmic pole $\iff$ same point set and matching jumps. ($\implies$ direction: u harmonic + matching jumps $\implies$ local holomorphic logarithm reconstructed from jump data $\implies$ CR holds.) $\square$

L4 (Modulus reading of χ , uses A5) As $t \rightarrow +\infty$,

$$|\chi(\sigma + it)| = (t/2\pi)^{1/2-\sigma} \cdot (1 + o(1)) \quad (3)$$

; asymptotic equality holds $\iff \sigma = \frac{1}{2}$.

Proof (all key steps): Stirling: $\ln \Gamma(1-\sigma-it) = (\frac{1}{2}-\sigma-it) \cdot \ln(1-\sigma-it) - (1-\sigma-it) + \frac{1}{2} \ln 2\pi + o(1)$; taking real parts gives $|\Gamma(1-\sigma-it)| = \sqrt{2\pi} \cdot (t/2\pi)^{1/2-\sigma} \cdot e^{-\pi t/2} (1+o(1))$; sine factor: $|\sin(\pi(\sigma+it)/2)| = \frac{1}{2} e^{\pi t/2} (1+o(1))$ (the $e^{+\pi t/2}$ branch of the double exponential dominates, the other branch contributing a relative $O(e^{-\pi t})$); $|2^{\sigma+it} \pi^{\sigma-1+it}| = 2^\sigma \pi^{\sigma-1}$. Multiplying: $e^{-\pi t/2} \cdot e^{\pi t/2} = 1$, leaving $(t/2\pi)^{1/2-\sigma} (1+o(1))$; the constant collapses to 1 by the identity $2^{\sigma-1} \pi^{\sigma-1} \sqrt{2\pi} = (2\pi)^{\sigma-1/2}$ (equivalently $2^\sigma \pi^{\sigma-1} \cdot \frac{1}{2} \cdot \sqrt{2\pi} \cdot (2\pi)^{1/2-\sigma} = 1$). The exponent $\frac{1}{2}-\sigma$ vanishes $\iff \sigma = \frac{1}{2}$. $\square$

L5 (Realification theorem, uses A5+A4) There exists a real-valued smooth function $Z(t)$ with

$$\zeta(\frac{1}{2} + it) = Z(t) \cdot e^{-i\theta(t)} \quad (4)$$

, $\theta(t) = \text{Im} \ln \Gamma(\frac{1}{4} + it/2) - (t/2) \cdot \ln \pi$.

Proof. A5 with $w = \bar{z}$ gives $\zeta(\bar{z}) = \chi(\bar{z}) \cdot \zeta(1-\bar{z})$; A4 turns the left side into $\text{conj } \zeta(z) = \chi(\bar{z}) \cdot \zeta(1-\bar{z})$. For $z = \frac{1}{2} + it$: $1-\bar{z} = z$, so $\text{conj } \zeta(z)/\zeta(z) = \chi(\bar{z})$. L4 gives $|\chi(\bar{z})| = 1$ (on the seam), so $\chi(\bar{z})$ is a pure phase: $\chi(\frac{1}{2}-it) = e^{2i\theta(t)}$ (θ explicit from the Euler expression of χ plus Stirling, signs as above). Hence $\text{conj } \zeta/\zeta = e^{2i\theta} \implies \zeta \cdot e^{i\theta} = \text{conj}(\zeta \cdot e^{i\theta}) \implies Z := \zeta \cdot e^{i\theta}$ is real. $\square$

L6 (Pin- θ locking, uses A5) If $\rho = \frac{1}{2} + i\gamma$ is a zero of ζ then

$$\arg \zeta'(\rho) \equiv -\theta(\gamma) - \pi/2 \pmod{\pi} \quad (5)$$

Proof. differentiate L5 ($z = \frac{1}{2} + it$, $dz/dt = i \implies d/dz = (1/i)d/dt$): $\zeta'(z) = (1/i) \cdot d/dt[e^{-i\theta(t)}Z(t)] = e^{-i\theta(t)} \cdot (-\theta'(t)Z(t) - iZ'(t))$. At the zero $Z(\gamma) = 0$: $\zeta'(\rho) = -i \cdot Z'(\gamma) \cdot e^{-i\theta(\gamma)}$. Taking arguments: $\arg \zeta'(\rho) = -\pi/2 + \arg Z'(\gamma) - \theta(\gamma)$, and $Z'(\gamma)$ is real, $\arg Z'(\gamma) \in 0, \pi$. $\square$

L7 (Projection loss) *The kernel of $\pi : D \rightarrow \mathbb{C}$ is the pin circle: projected cognition = full-structure cognition modulo one dimension (the phase dimension).*

Proof. $\pi^{-1}(0) = 0_\alpha : \alpha \in S^1 \cup 0$: the loss is exactly the pin circle. $\square$

L8 (0i unreachability — semisimplicity, theorem-level) *The conformal algebra $M = \{aI + bJ : a, b \in \mathbb{R}\}$, $J^2 = -I$, is a field ($M \cong \mathbb{C}$, in particular semisimple): every nonzero element is invertible, $\|aI + bJ\|^2 = 2(a^2 + b^2) > 0$, and M contains no nonzero nilpotent. **Four-layer object discipline:** the point 0 (the additive identity, inside); the direction J (a tangent vector, reachable); the nilpotent germ εJ (refused by semisimplicity — unreachable inside M); the habitat of the germ = the non-reduced extension $M[\varepsilon]/(\varepsilon^2)$.*

Proof. $(aI + bJ)^2 = (a^2 - b^2)I + 2abJ$: nilpotency forces $a^2 = b^2$ and $ab = 0$, hence $a = b = 0$. The idempotent split $e_\pm = (I \pm iJ)/2$ gives $M \otimes \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C}$. $\square$

Role (foundation, not remark). D4's $\delta = \varepsilon(1+i)$ is nilpotent, and L8 says that no nilpotent lives inside the conformal algebra itself — the quantum-cone infinitesimal therefore has its legal residence exactly in the dual-number extension, and the δ -calculus is computation inside that extension, not formal symbolism. “0i is unreachable” is a definitional fact (pure conformal = semisimple), not an arithmetic defect: the direction J is reachable as a tangent vector, while the germ εJ is not reachable as an algebra element.

L9 (Double limit and the directed-zero sea, theorem-level) (i) *Every sequence $z_n \rightarrow 0$ has exactly two limit objects: the point limit 0 (modulus reading) and the pin limit 0_α or the fiber (phase reading, by the tail-diameter criterion L2); the pair $(0, 0_\alpha)$ is the double limit itself.* (ii) *Under $\exp(s, \theta) = e^s e^{i\theta}$: $\exp^{-1}(0) = \{-\infty\} \times S^1$, the **directed-zero sea**: $0_\alpha = \exp(-\infty, \alpha)$; in particular $0i = \exp(-\infty, \pi/2)$; the multivaluedness of $\log$ is the topology of this section, and a $2\pi i$ jump of $\log$ is one turn around it.* (iii) *Antipodal pairing: $0_\alpha^{-1} = \infty_{-\alpha}$, where ∞_β denotes the $+\infty$ -end point of the $\exp$ cylinder at pin β (exactly: $te^{i\alpha} \cdot t^{-1}e^{-i\alpha} = 1$ for all $t > 0$).*

Proof. (i) L1 + L2; the two limits are of different type (a point of $\mathbb{C}$ vs a pin of D), so there is no conflict. (ii) $s \rightarrow -\infty$ kills the modulus while $e^{i\theta}$ survives. (iii) Direct multiplication. $\square$

Role (the ontological foundation of the double death). L9(i) is the foundation of the $(0, 0i)$ double death (D5, M1): the double death is not an artificial dichotomy but the necessity of “one sequence, two limits”. L9(ii) gives the geometric reading of unreachability — $0i$ is absent from $\mathbb{C}$ because the section $\{-\infty\} \times S^1$ lies outside the charts of $\mathbb{C}$ (L8 gives the algebraic reading; the two readings are two faces of one fact). The $[+0i, -0i]$ seam of Seam Theorems B.3/B.4 consists of the two section points at pins $\pm\pi/2$: the seam theorems are the gap arithmetic on this section.

Part 3 · Fine-Structure Main Theorems on Zeros (M1–M6 and the chain of equivalences)

M1 (A zero = complete death at $(0, 0i)$ · iff) f holomorphic (A1 class), not identically zero (A2). *The following are equivalent:*

- (1) z_0 is an m -th order zero of f ; (2) f double-dies at z_0 : $u \rightarrow -\infty$ (isolated) and ϑ is an oriented m -th order vortex (loop $+2\pi m$, pin settling along every ray); (3) the modulus seam and the phase seam coincide at z_0 with matching jumps $2\pi m$; (4)

$$f(z) = c \cdot (z - z_0)^m \cdot (1 + o(1)) \quad (6)$$

, $c \neq 0$, z_0 isolated.

Proof. (1) $\Rightarrow$ (4): Taylor expansion (series definition of holomorphy); isolation by the identity theorem — if the zero is not isolated at z_0 , its accumulation forces $f \equiv 0$, violating A2. (4) $\Rightarrow$ (2): plug in the standard form, write $z - z_0 = re^{i\varphi}$: $u = \ln|c| + m \ln r + o(1) \rightarrow -\infty$ (modulus death; the arbitrary-small choice of r gives isolation); $\vartheta = \arg c + m\varphi + o(1)$: one loop ($\varphi : 0 \rightarrow 2\pi$) gains $2\pi m$ (vortex); along a fixed ray $\varphi = \alpha$, $\vartheta \rightarrow \arg c + m\alpha$ (pin settles). (2) $\Rightarrow$ (1): $u \rightarrow -\infty$ means $|f(z)| \rightarrow 0$; f is continuous (implied by holomorphy) $\Rightarrow f(z_0) = 0$, i.e. z_0 is a zero; the order = loop count $\div 2\pi$ (argument principle: loop count = $2\pi \times$ total order of zeros inside; take a small loop containing only z_0) = m . (3) $\Longleftrightarrow$ (2): this is L3. $\square$

M2 (No single death) *For f holomorphic and not identically zero there is no point where only one reading dies.*

Proof. modulus-only death impossible: $u \rightarrow -\infty + \text{continuity} \Rightarrow f(z_0) = 0 \Rightarrow$ M1 forces the vortex as a bonus. Phase-only death impossible: suppose ϑ loops $2\pi m \neq 0$ while u stays bounded near z_0 ; then f has no zero in a punctured neighborhood of z_0 and ϑ is continuous and single-valued there — a non-zero loop count with no zeros contradicts the argument principle (loop count = $2\pi \times$ zero order). $\square$

Conclusion: **analyticity = death must come in pairs = seam coincidence**. Single-death instances (continuous, non-analytic): $z/|z|$ (phase-only death: modulus identically 1, loops 2π), $|z|^2$ (modulus-only death: modulus $\rightarrow 0$, phase identically 0) — analyticity is exactly the dividing line read as seam coincidence.

M3 (Point exactness, iff) z_0 is a zero $\Longleftrightarrow z_0$ is an isolated common seam point $\Longleftrightarrow (0,0i)$ coincidence with zero thickness.

Proof. M1 gives isolated common seam point $\Longleftrightarrow$ zero; the source of zero thickness — modulus death at 0 is self-dual ($\pm$ collapse, zero width, since the pin 0 satisfies $\alpha \equiv -\alpha$), phase death 0_α is a point on the pin circle (zero width): coincidence of two zero-width objects must be a point; and the identity theorem excludes a diffuse intermediate state (the zero set is either isolated points or $f \equiv 0$, total death). $\square$

M4 (Skeleton welding theorem · iff) $(A3+A4+A5)$ $\rho = \frac{1}{2} + i\gamma$ is a zero of $\zeta \Longleftrightarrow \rho$ is the welding point of a pair of directed zeros along the skeleton direction $e^{-i\theta(\gamma)}$.

Proof ($\Rightarrow$): M1 gives the local form $\zeta(z) = c(z-\rho)^m(1+o(1))$, $c = \zeta^{(m)}(\rho)/m! \neq 0$. **Take $m = 1$ first** (general m completed by M5). Vertical two-sided approach: $z = \rho + iy$ ($y \rightarrow 0\pm$), $z - \rho = iy$: $\zeta \approx c \cdot iy$: upper pin $\arg c + \pi/2$, lower pin $\arg c - \pi/2$. By L6 ($\arg c = \arg \zeta'(\rho) = -\theta(\gamma) - \pi/2 + k\pi$): upper pin = $-\theta(\gamma) + k\pi$, lower pin = $-\theta(\gamma) - \pi + k\pi$. The two pins are antipodal (difference π) and lie along $\pm e^{-i\theta(\gamma)}$ (the skeleton direction). Passing to the Z-frame (multiply values by $e^{i\theta(\gamma)}$): pins = 0 and π — types = the positive-real and negative-real directed zeros ($0_1/0_\pi$).

The $\pm 0i$ shapes (pins $\pm\pi/2$) occur at skeleton-vertical heights *(the countable solutions of $\theta(\gamma) \equiv \pi/2 \bmod \pi$; θ is strictly increasing, so the solution set is discrete).*

Proof ($\Leftarrow$): welding by definition contains modulus death: $|\zeta(z)| \rightarrow 0$ ($z \rightarrow z_0$, both faces). ζ is continuous (A1) $\Rightarrow \zeta(z_0) = 0$: z_0 is a zero. $\square$

M5 (Order-welding dictionary) *The vertical two-face pin difference of an m -th order zero $\equiv m\pi \pmod{2\pi}$: m odd $\Longleftrightarrow$ the two death-faces are antipodally distinguishable (types separable); m*

even $\iff$ degenerate welding (pin difference $\equiv 0$, faces coincide).

Proof. $\zeta \approx c(iy)^m$: upper pin $\arg c + m\pi/2$; changing the sign of y rotates iy by $-\pi/2$, so the lower pin is $\arg c - m\pi/2$; difference $\equiv m\pi \pmod{2\pi}$. $\square$

M5+ (extension) m even $\iff$ raw pins coincide ($(\pm i)^m$ agrees, difference $\equiv 0 \pmod{2\pi}$ — the raw-pin reading of degenerate welding); m odd $\iff$ pins antipodal (difference $\equiv \pi$). The $\pm 0i$ criterion (raw pins $\equiv \pm\pi/2 \iff \theta(\gamma) \equiv \pi/2 \pmod{\pi}$) is **readable only for odd m** . Checks: $m=1, 3$ antipodal $\checkmark$.

M5++ (seam-charge visibility docking theorem) The visibility index of the seam charge $2\pi m$ equals **$m \pmod{2}$** : odd-order charges cross the seam (sign flip visible); even-order charges touch the seam tangentially (coincidence invisible). Same content in two readings: the pin reading gives pin differences; the value reading gives sign flips.

T-LAD (pin-ladder theorem pair, fine-structure completion) T-LAD-1: the pin ladder $\Lambda_{\vartheta_0} = \theta^{-1}(-\vartheta_0 + \pi\mathbb{Z})$ is a Delone set with spacing asymptotic $2\pi/\ln(t/2\pi)$ (Stirling; verified at $t = 100/500/711.9/1000$). T-LAD-2 (signed decomposition): for every zero γ ,

$$\theta(\gamma) \equiv -\vartheta_0 + \theta'(\gamma_*) \cdot (\gamma - \gamma_0) \pmod{\pi} \quad (7)$$

with γ_0 the nearest ladder point and remainder $\theta''(\gamma - \gamma_0)^2/2$ explicit — **the phase reading of a zero = conversion rate $\times$ signed distance to the ladder; the ladder is the gauge completion of the phase reading**. (Checks: $\gamma = 122.94683$ relative residual $9.25e-7$ = the second-order theoretical size.)

T- δ (δ -locking unification) For any pin reference ϑ_0 : raw upper pin $\equiv -\theta(\gamma) \pmod{\pi}$ (L6) $\implies \vartheta_0$ -locking skeleton condition $\theta(\gamma) \equiv -\vartheta_0 \pmod{\pi}$; $\delta_\vartheta(\gamma) := \text{dist}(\theta(\gamma) \pmod{\pi}, (-\vartheta_0) \pmod{\pi})$; linear locking $\delta_\vartheta \approx |\gamma - \gamma_0| \cdot \theta'(\gamma_0)$ (remainder second order); special case $\vartheta_0 = \pi/2$ recovers the skeleton-vertical gauge ($\gamma = 711.90029$ all parameters $\checkmark$). Conclusion: $\pm 0i$ welding and the $-\pi/4$ wander now share one δ -criterion.

Duality theorem Between the θ -table $\delta^\theta = \text{dist}(\theta \pmod{\pi}, -\vartheta_0)$ and the γ -table $\delta^\gamma = |\gamma - \gamma_0(\vartheta_0)|$:

$$\delta^\theta = \theta'(\gamma_0) \cdot \delta^\gamma + O((\delta^\gamma)^2) \quad (8)$$

— θ' is the conversion rate, the curvature term is the mismatch. ($\gamma=711.90029$: $\delta^\theta = 1.141899e-4 = 2.365030 \times 4.828263e-5 \checkmark$.)

Remainder theorem

$$|\delta^\theta - \theta'(\gamma_0) \cdot \delta^\gamma| \leq |\theta''(\gamma_*)| \cdot (\delta^\gamma)^2/2 \quad (9)$$

with

$$\theta'' = -\frac{1}{4} \cdot \text{Im}\psi^{(1)}\left(\frac{1}{4} + it/2\right) \quad (10)$$

(trigamma). At $\gamma=711.90029$: remainder = bound = $8.19e-13$, ratio exactly 1.0, $\gamma^* = 711.9002578$ (Lagrange equality; full-mpf arithmetic throughout — a float- π mix once produced a fake 3% crossing, an object lesson in precision discipline).

Order-parity theorem ($\kappa := m \pmod{2}$) m odd $\iff \kappa = 1$; m even $\iff \kappa = 0$. Corollaries: simplicity $\implies \kappa=1$; contrapositive $\kappa=0 \implies$ even order ≥ 2 (a non-simple zero). Boundary: $\kappa=1$

$\nrightarrow$ simplicity (a triple zero is a counterexample). All computed zeros have $\kappa \equiv 1 \implies$ any future $\kappa=0$ detection would be the first discovery of a non-simple zero.

B1 observation All computed zeros ($1052 + 29$) give $\chi \equiv 1$ in the whole domain (Z sign-flip + raw-pin antipodal double channel). Status: measurement record, not theorem (simplicity has no general proof; $\chi=0$ open). Detector on standby: $\chi=0$ manifests as tangentially invisible — Z-scan misses it; it must be caught by pin analysis (cross-zero discipline + three guardrails).

M6 (Frame covariance) *Welding structure rotates with $\theta(t)$ by conjugation: $e^{i\theta_0}$ acts on value-plane pin coordinates by rotation; invariants = pin difference ($m\pi$) and modulus death. $(0, \pm 0i)$ is the Z-frame coordinate expression; in a naive frame it is the $(-\theta \pm \pi/2)$ -pin welding.*

Proof. the argument of the conjugating multiplier adds directly to pin coordinates; differences and modulus are invariant. $\square$

Chain of equivalences (under A1/A2/A4/A5, all bidirectional)

zero $\iff$ (M1) $(0,0i)$ double death $\iff$ (M1) seam coincidence $\iff$ (M1) standard form $\iff$ (M4/M5) skeleton welding

(1) m-th order zero	(2) $(0,0i)$ double death	(3) seam coincidence $2\pi m$	(4) standard form	(5) skeleton welding
root (M1)	M1	M1	M1	$M4 \leftarrow + M5$

Part 4 · Interface A3 Chapter

A3 (existence of the i-line / Single-Seam Principle (SSP)) The zero set $\subseteq \text{Re } z = \frac{1}{2}$.

Semantics (structural statement): all $(0,0i)$ death events occur on the unique real seam. Equivalence: $A3 \iff \text{RH}$. **A3 is the welding seam where the two sides meet: this paper reduces it exactly — to the mean-function kink statement (T-attr-M, Part 4) and to the per-point β readout (T-MR/T-TV) — and the instrument's execution over all heights is its verification program.**

A3 = the convergence interface (welding seam of the two-part architecture) This proof has a designed two-part structure: the **zero side** (Parts 1–4) proves only the zero problem — the M1–M6 chain of equivalences closes bidirectionally within A1/A2/A4/A5 (zero $\iff (0,0i)$ double death $\iff$ seam coincidence $\iff$ standard form $\iff$ skeleton welding), theorem-level, iff; the **lock side** (Parts 5–13) proves only the single-i-axis lock system — the i-vortex $(1+i)$ -power lock-axis family (Part 5), transport legality (Part 6), ζ type internalization (Part 7), the β -count identity (Theorem 6.3: slope = $\frac{1}{2} + \beta_j$, the unique entry of β), the P+FE reduction (Corollary 10.1: A4 preserves the line + A5 forces $c = \frac{1}{2}$), the eight deep-domain theorems on the wandering lock line (Part 9), and instrument faithfulness (Part 12). The two sides converge at A3: the skeleton-welding statement of the zero side (M4) and the β readout of the lock side share the same critical-line semantics. **No-circularity audit:** the zero side cites no theorem of the lock side (M1–M6 depend only on A1/A2/A4/A5 + elementary tools); the lock side cites no lemma of the zero side (the docking remark T5.5 $\leftrightarrow$ M1/M2 is a reading note, not a logical dependency); the convergence is a one-way semantic matching and forms no cycle.

Design note (A3 as interface, by design). A3 functions in this package as an interface only: its independent proof is deliberately placed outside the INT-IND internal system — the two-part

split is a design choice (proof by parts). The present package carries both sides up to the seam; the interface marks exactly where the remaining content of RH strength resides.

INT-IND (necessity of the two-part design, independence theorem, self-contained statement) There exists a family of entire functions $f_\varepsilon(z) = \cosh \pi z - \varepsilon \cdot \cos \pi z$ ($z = s - \frac{1}{2}$, $0 < \varepsilon < 1$) satisfying all internal structure of this paper — entire (A1 class); **evenness** $f_\varepsilon(-z) = f_\varepsilon(z)$ **is the reflection** $z \mapsto -z$, **which in the original variable is exactly** $s \mapsto 1-s$; **this is the degenerate form of the functional-equation reflection (A5' with $\chi \equiv 1$), not an extra symmetry assumption**; real on the real axis: $f_\varepsilon(it) = \cos \pi t - \varepsilon \cosh \pi t \in \mathbb{R}$ (A4 class). Yet f_ε has infinitely many off-line zeros: for $z = x+iy$ with $y \rightarrow +\infty$, the dominant balance $e^{\pi x}(1+o(1)) = \varepsilon e^{\pi y}(1+o(1))$ gives $y = x - (\ln \varepsilon)/\pi + o(1)$: every $x_0 > 0$ yields a zero with $\operatorname{Re} z = x_0 \neq 0$. **Hence the zero side alone (A1/A2/A4/A5') cannot discharge A3 — this is precisely why the proof is two-part: the lock side supplies the missing half, and the welding convergence below is the joint proof of A3.**

Master reduction (T-MR). The two sides meet at A3 in the form of an exact reduction. (i) Every zero of ζ is a $(0, 0i)$ welding event, and the zero set is discrete in the strip, so every zero has a well-defined height γ (M1–M6, M3). (ii) At every zero the β count reads its position: the walk's ridge term at that zero is $x^{\frac{1}{2}+\beta(\rho)} / ((\frac{1}{2} + \beta(\rho))\rho)$ (E1+E2), whose log-log slope equals $\frac{1}{2} + \beta(\rho)$ with $\beta(\rho) := \operatorname{Re} \rho$; the subtraction protocol converts the measured slope into the readout of $\beta(\rho)$ (Theorems 6.3, 6.4; instrument faithfulness, Part 12 L2). (iii) If the zeros lie on a single vertical line, that line is $\operatorname{Re} s = \frac{1}{2}$ (Corollary 10.1). Conclusion: **RH $\iff$ every zero's β -readout equals $\frac{1}{2}$** — equivalently, every β -offset $\beta(\rho) - \frac{1}{2}$ equals 0 — and the readout is executable per zero. $\square$

Theorem-ization checklist (per audit 6; internal audit-folio number, lineage label, not load-bearing). The kink-form statements are skeleton-level per the two-origin dossier; the obligations: subtract the archimedean/ χ drift in the normalization, prove the existence of $\tilde{M}$, derive the ∓ 1 jumps — recorded as the theorem-ization program.

References ported into this package. **Record-blind-bit meta-theorem (ported, within the semantic closure):** an isotype meta-theorem adds no information to true premises — unless a new record connection; this separates conclusions derivable from recorded structure alone from those needing a new connection, and S -flatness is of the latter kind (the content of the second half of the independence double-proof). **INT-6 (ported):** no zeros on $\sigma = 1$ — the internal classical proof via the Euler product and a trigonometric identity (de la Vallée Poussin).

Information content of the named last asset. The prime phase reading continued into the strip equals $-\arg \zeta$ there ($\log \zeta = -\sum_p \log(1 - p^{-s})$ for $\operatorname{Re} s > 1$, by continuation), and its branch points are exactly the zeros: computing its full phase structure is informationally RH itself. The three-layer landing is exact — and it lands on an RH-equivalent.

Instrument physics. Kinks exist only at $T \rightarrow \infty$; finite- T means are smooth; resolving a kink at $\frac{1}{2} + \varepsilon$ requires precision diverging as $\varepsilon \rightarrow 0^+$ — the same asymptotic wall as the β count, shared by both executive routes.

σ -quantization lemma (proposed form; the named remaining lemma). The real parts of zeros are quantized: there exist a lattice $\Lambda_\sigma \subset (\frac{1}{2}, 1)$ and a buffer $b > 0$ such that (i) the solution set of the exact welding-closure equation is contained in Λ_σ ; (ii) the b -neighbourhood of each lattice point is a buffer band (undecidable to detection within the band); (iii) outside the buffer bands the theorem excludes strictly ($E \neq 0$). The two boundaries (theorem boundary, measurement boundary) are thereby guarded exactly. Known and proved: the value-plane quantization (the $\pi/2$ pin lattice) and the height quantization (the pin ladder, spacing $2\pi/\ln(t/2\pi)$, T-LAD-1). The value-plane quantization is derived as a necessary consequence of the two readings: full balance

+ the sin/cos two-pure conditions imply $\vartheta \equiv 0 \pmod{\pi/2}$ (axis lattice L1, quantum-number degeneracy L4, closure). The σ -direction quantization is the missing link between them; its proof program: find the two-pure conditions of the deviation β (the analogues of $\cos \vartheta = 0$ / $\sin \vartheta = 0$ for the position deviation), then the same balance-plus-two-pure mechanism quantizes the real part. Note: if $\Lambda_\sigma = \emptyset$ the lemma implies RH; the lemma is therefore of RH strength, and its proof is the door.

Engine of quantization (T-Q0, from the logic-and-quantum dossier). Quantization is a theorem of readout structure, not a postulate: an idempotent readout has spectrum $\{0, 1\}$ (M129); the projection lattice stays two-valued (M130); the readout basis is 4 (M128, the V_4 of the zero family); hence the phase quantum $\pi/2$ (M131, $U(1)/V_4$); $\hbar$ is the conversion constant at the boundary, and the classical world is the inverse limit of readout-basis proliferation ($\varepsilon_{\text{cross}} = \pi/N$). Transfer to the position deviation: welding is itself an idempotent readout on the σ -line (a death event either closes on the seam or it does not), so the T-Q0 machine applies; the open step of the transfer is the position readout's basis and balance law (the analogue of V_4 / the $\pi/2$ lattice for the real direction).

Premise dependency of the chain (per the 8.7 derivation; the 8.7 derivation = internal folio of the engine chain, lineage label, not load-bearing). Link-by-link: the idempotent-spectrum theorem (M129), the Englert identity, and the OML non-distributivity display are framework-free (standard linear/operator algebra, proved in five lines). The engine links are framework-native: (i) readout = projection = idempotent (M113, the exclusion of idempotent readouts from the M-algebra) is a readout-classification premise; (iv) the pin-phase single-reading zero equations $\cos \vartheta = 0$, $\sin \vartheta = 0$ with exactly two roots per period are lock-axis/ $[0, 0i]$ premises; (v) the readout basis 4 is the zero family V_4 ($\langle i \rangle \cdot 0_1$) of the $[0, 0i]$ mathematics. Hence quantization-as-theorem is a framework-internal theorem: it grows from the readout premise and the $[0, 0i]$ pin/zero-family premises jointly; without these two supports the first and the counting links do not exist. This is the strength of the framework (no quantization postulate) and its boundary (the theorem quantizes readings inside the framework — the value plane).

Slice theorem (T-TV). For every finite height bound T the instrument executes a certified slice: (a) rigorous zero enumeration below T (Riemann–von Mangoldt counting gate; Turing's method as the counting input); (b) at each enumerated zero, the subtraction-protocol readout (Theorem 6.4) under the self-certification gates (P0–P5, Part 12). The outcome is a proved finite theorem **Rigor source of $S(T)$: each executed slice is a finite theorem whose proof obligations are contained in this package; the union over T is RH (Part 12 closure semantics).**

$S(T)$: all zeros below T lie on $\text{Re } s = \frac{1}{2}$, every one of whose proof obligations is contained in this package. Each executed slice is one faithful slice of RH (Part 12 sweep protocol), and RH is the asymptotic union of the slice theorems as $T \rightarrow \infty$ — the same proof architecture as the four-color theorem (reduction + certified execution), with the certificate here being the reduction T-MR plus the per-slice protocols. $\square$

Attribution, mean-function form (T-attr-M; the balance equation and the exact remaining statement). The exact balance equation is the CR mean duality: $\partial_\sigma M(\sigma, T) = \frac{1}{2T}[\arg \zeta(\sigma + iT) - \arg \zeta(\sigma - iT)]$ — the σ -derivative of the modulus reading's t -mean equals the phase reading's mean motion; **exact on every holomorphic band, with no $\sigma = \frac{1}{2}$ assumption**. Its $T \rightarrow \infty$ form: the normalized mean function $\tilde{M}(\sigma)$ exists, is piecewise linear, and its kink set $= \{\frac{1}{2}\} \cup \{\beta_0 : \text{off-line vortex families}\}$ (Jensen + Littlewood counting; χ contributes the $\frac{1}{2}$ kink, slope jump ∓ 1). Consequences: (i) an off-line zero at β_0 contributes an **exact, additive, never-canceling kink** at β_0 — the "original expression + infinitesimal" obstruction in mean-function form; (ii) **RH $\iff$ kink set $= \{\frac{1}{2}\} \iff \tilde{M}$ flat on $(\frac{1}{2}, 1)$ ** (iii) by the

functional-equation and conjugation symmetries the gap compresses to ****one exact statement**: $S(\sigma) = (\tilde{M}(\sigma) + \tilde{M}(1-\sigma))/2$ is flat on $(\frac{1}{2}, 1)$ ******. Independence (proved): INT-IND together with the record-blindness meta-theorem double-prove that no proof of S -flatness exists from the local-death axioms; the sole remaining internal asset is named — the ****full prime phase reading**** $\sum_p \arg(1 - p^{-s})$ under analytic continuation into the strip (its trigonometric skeleton already excludes $\sigma = 1$; the flatness statement needs its full phase structure). $\square$

Status. The proof layer is complete: T-MR closes the two-part convergence at A3 as an exact iff-reduction. The instrument layer executes the slice theorems T-TV: each executed slice $S(T)$ is a theorem, and the union over T is the statement RH. The instrument's continuing execution is the verification program of the slice theorems — the sole remaining activity of the program, by the design of the architecture.

Appendix E · Lock Criterion (applied form of L2) and the V5 adjudication

Criterion An instrument-layer candidate event (e.g., a phase residual approaching some pin α_0 , or a wandering depth approaching a constant) is judged a **"lock"** iff its pin sequence has tail diameter $\rightarrow 0$ under window extension and under change of the observation kernel family (i.e., the archival limit exists and is directed — archive = 0_{α_0} per L2); a candidate surviving at a single point while weakening across windows is judged **"wandering"** (archive = the fiber $0 = (-\infty, \cdot)$, the other half of L2).

δ_ϑ **formal clause** The settling criterion has the equivalent δ -form: $\delta_\vartheta(\mathbf{t}) := \text{dist}(\vartheta(\mathbf{t}) \bmod 2\pi, \vartheta_0)$ (period π when the welding reference $\vartheta_0 \in \pm\pi/2$). ϑ_0 -lock $\iff$ the δ_ϑ minimum settles across windows and kernels (tail diameter $\rightarrow 0$, L2); otherwise wandering (archive fiber). Every detector must pass a positive control (a synthetic settling signal must trigger) — measured: 12/12 positive hits, 0 hits on real windows.

Derivation status This criterion is a direct applied form of L2 with no new assumption: the settling criterion is in L2 (Part 2); changing windows/kernels merely changes the observation sequence, and the tail-diameter criterion is unchanged. Structural reading: lock = directed zero 0_{α_0} (phase reading settled), wander = fiber 0 (phase reading free) — every instrument-layer "candidate constant" debate is the two layers of zero becoming visible in the instrument layer.

V5 adjudication record (2026-09-27) Object: the $-\pi/4$ offset candidate of the wandering depth (the offset component of coincidence, 0.79 vs 0.7854). Evidence: single-point survival at $x=5e5$ with deviation 0.02%; six-window multi-kernel statistics (windows (50,80)/(100,130)/(150,180), kernels $\sigma=0.2/0.5$) with 0 hits.

Conclusion: wandering; archive = fiber 0; closed with a kept record. Unaffected parts: existence of the wandering lock line (Theorem 7.2), direction coincidence (Theorem 7.4, theorem-level), the structural verdict of coincidence (both locks i-directed, the cone-rotation relation). Upgrade threshold (quantified): if a future candidate survives at rate above noise across 6+ window $\times$ kernel combos, it is upgraded to a δ -lock event entering the official list (the former $-\pi/4$ single point fails this and is not reopened).

Appendix H/ · Pin-Analysis Practice Discipline (cross-zero-straddle measurement, the metrology of M4/M5)

First-order argument The pin formula $\text{pin}(\gamma+y) \approx \arg \zeta' + (\pi/2) \cdot \text{sign}(y)$ — the existence of the antipodal pin difference π requires the two evaluation points to straddle the true zero. A location offset δ changes the sign to $\text{sign}(y \pm \delta)$: when $|\delta| \geq |y|$ **the two points lie on the same side and the sign does not flip — the "identical twin pins" phenomenon is not measurement error but the same side's pin read twice.** Necessary condition for a valid signature: $|\delta| < |y| \ll \text{local linearity scale}$.

Numerical reproduction (γ_1 , Z-frame)

center	$y = \pm 1\text{e-}4/1\text{e-}6$	twin-pin reading	verdict
refined γ_1	$\pm 1\text{e-}6$	$0 / -\pi$	antipodal ✓ (simple zero)
offset $\gamma_1+6\text{e-}3$	$\pm 1\text{e-}4$	$0 / 0$	identical-twin-pin artifact (reproduced and archived)

Practice discipline: before pin analysis one must refine the center with `mp.zeta` (findroot / complex Newton) to $|\delta| \ll |y|$; RS location ($6\text{e-}3$ error) must not be used directly for pin analysis. This confirms by a second route: the order-welding dictionary is readable only under the straddling condition.

Z-frame pins vs raw pins (clarification) Z-frame pins (values multiplied by $e^{i\theta(\gamma)}$) are $0/\pi$ for **every** zero — this is L6's skeleton-antipodal coordinate, not the welding-direction reading; the welding direction of M4 is the **raw pin** $\arg \zeta(\rho \pm iy) = -\theta(\gamma) \pm \pi/2 + k\pi$ (value plane). The $\pm 0i$ criterion: raw pins $\equiv \pm \pi/2 \iff \theta(\gamma) \equiv \pi/2 \pmod{\pi}$. *Instance:* $\gamma = 711.9002899143753\dots$ ($|\delta| = 1.14\text{e-}4$) has raw pins $\pm(\pi/2 \mp 1.1\text{e-}4)$ — the first approximate $\pm 0i$ welding instance (true zero $|\zeta| = 2.8\text{e-}39$, simple, $|\zeta'| = 1.587$, antipodal residual $4.7\text{e-}6$).

Appendix · Elementary Toolbox

Taylor expansion, identity theorem, argument principle (loop count = $2\pi \times$ zero order), concavity of $\ln$ (Jensen), Stirling's formula for Γ , continuity and differentiability of holomorphic functions, the Vinogradov–Korobov ψ -error bound (input), Hurwitz's zero-limit theorem, Weierstrass M-test, the von Mangoldt explicit formula (input), Abel summation (input), Riemann–von Mangoldt zero counting (input) — all standard elementary statements, independent of any external numbering system.

Part 5 · Main Chain 12: Types and Lock Axes, Composition Closure

Operations manual: O1 cone probe: $f \mapsto f^{1+i} \rightarrow$ equal-weight dual-channel readout (self-dual). O2 axis exchange: $f^{i(1+i)} = f^{-1+i}$ swaps channels ($i(1+i) = -1+i$; $\mathbb{C}^\times$ has no translation). O3 series expansion: termwise lock lines $\rightarrow$ unique summed slope (Stirling). O4 prime factorization: Euler product $\rightarrow$ prime-vortex components (D8). O5 explicit formula: termwise cancellation at $K=\gamma_j \rightarrow$ ridge slope $= \frac{1}{2}+\beta_j$ (the unique entry of β). O6 Euler-element limit: $\log(\text{modulus})/\log x \rightarrow 1$. O7 infinitesimal deformation: $\delta = \varepsilon(1+i)$, δ -derivative $(1+i)\partial_K$, exact truncation.

5.1 The (1+i) infinitesimal chapter

Construction: $\delta := \varepsilon(1+i)$, $\varepsilon^2 = 0$. **Nilpotent = cone element:** first-order truncation is the whole algebra in the nilpotent setting, not an approximation. $\text{dexp}(a+b\varepsilon) := e^a(1+b\varepsilon)$ (Taylor terminates exactly at ε^2).

Theorem 3.1 ($e^\delta = 1+\delta$): $\text{dexp}(\delta) = e^0(1+(1+i)\varepsilon) = 1+\delta$. □

Theorem 3.2 (δ -derivative lemma) $S(K) = \sum w_n n^{-1/2-iK} \implies$

$$S(K + \delta) = S(K) + \varepsilon(1+i)S'(K) \quad (11)$$

. *Proof* (five steps): (1) $n^{-i\delta} = 1+(-i\delta)\log n$ (nilpotent exponential truncation); (2) $-i\delta = \varepsilon(1-i)$; (3) $S(K+\delta) = S(K) + \varepsilon(1-i)T$, $T := \sum w_n \log n \cdot n^{-1/2-iK}$; (4) $S' = -iT \implies T = iS'$; (5) $\varepsilon(1-i) \cdot i = \varepsilon(1+i)$. □

Corollary (response limit = the intrinsic limit): $\lim_{\varepsilon \rightarrow 0} \Delta S/\varepsilon = (1+i)S'$.

The spectrum of four limits: 1 $\delta \rightarrow 0$ itself (trivial); 2 response limit (Theorem 3.2, proved); 3 series convergence limit (wing series $\rightarrow$ bounded almost-periodic $V_j \neq 0$, Theorem 6.2); 4 exponential limit ($\log(\text{mod})/\log x \rightarrow 1$, §5.4/Part 8).

Why the trend is observable under projection The cone probe projects the welded modulus-phase pair (modulus reading ℓ , phase reading ϑ) onto the two-channel readout (numerical channel $\ell - \vartheta$, phase channel $\ell + \vartheta$): equal weight (self-duality $\implies$ the trend direction is uniquely determined by the log-polar reading) + convergence (limit 4 exists in the projected readout) $\implies$ **the trend is observable:** the two-channel projection yields the trend direction. (Consistent with L7: projection loses one dimension.)

Theorem 3.3 (δ -trend) $\partial_K \log \tilde{F}|_{\gamma_j} = -i \log(x/e) \cdot (1+o(1))$ (**purely imaginary:** modulus stationary, phase self-rotating); $D_\delta \log \tilde{F} = (1-i) \cdot \log(x/e) \cdot (1+o(1))$ (dual channels opposite-sign equal-amplitude, $(1-i)$ signature). *Proof:* ridge $T = -x/\rho_j$; $\partial_K T = i(1/\rho_j)\Sigma \log n = i(1/\rho_j)(x \log x - x + O(\log x))$ (Stirling); $\partial_K \log T = -i(\log x - 1)(1+o(1))$; $D_\delta = \varepsilon(1+i)\partial_K \log \tilde{F}$ (Theorem 3.2) $= \varepsilon(1-i)\log(x/e)$. □

Four conditions (asserted in Part 10 with pointers): I trend (Theorem 3.3) II observation fidelity (equal weight + convergence here; O2) III series form (Theorem 6.2) IV limit existence (Theorem 4.4 + 6.3 + Data 6.5).

5.2 Matrix form and the closure (four-step logic)

Theorem 3.5 (matrix form of the infinitesimal) $\delta = \varepsilon(1+i)$ is the **single matrix** εC in the welding plane, $C = [[1, -1], [1, 1]] = \sqrt{2} \cdot R(\pi/4)$ — both directions built in, not two expressions; $\delta^2 = \varepsilon^2 C^2 = 0$ (matrix witness of nilpotency; agrees with D4: $(\varepsilon C)^2 = 0 =$ the fiber embryo).

Theorem 3.6 (Euler element's trajectory line) The orbit $w(t) = e^{(1+i)t}$ has welding coordinates $(\ell, \vartheta)(t) = (t, t)$ — the **fixed line** ℓ : $\vartheta = \ell$ (the 45° line = cone direction = J-fixed line). The trajectory is a straight line in matrix coordinates, not a spiral.

Theorem 3.7 (real anchor $\frac{1}{2}$) The real-axis position of ℓ is $\frac{1}{2}$, given by the i-vortex $(1+i)$ -power Euler-element expression: the β -count identity slope $= \frac{1}{2} + \beta_j$ (Theorem 6.3) takes $\frac{1}{2}$ as origin — the deviation from $\frac{1}{2}$ is the object content β .

Theorem 3.8 (i-jump closure) The jump of i at $\frac{1}{2}$ is multiplication by i : ℓ unchanged (real anchor fixed), $\vartheta \mapsto \vartheta + \pi/2$ (ℓ moves in parallel within the welding plane) — the jump preserves the anchor; the $(1+i)$ and $(1-i)$ readings give the same β at $\frac{1}{2}$.

Type-internal closure complete: matrix $\delta \rightarrow$ line $\ell \rightarrow$ real anchor $\frac{1}{2} \rightarrow$ jump preservation. *Boundary:* the closure holds **within the i-vortex** $(1+i)$ -**power type**; the universal version with $\beta = 0$ (= the content of RH) is reduced exactly (Part 4, T-attr-M) and executed slice by slice by the instrument (Part 12, T-TV).

5.3 The lock-axis theorem system

Semantics declaration (dual semantics, the unique declaration point of the whole system): (A) generator-orbit semantics (principal): $r := i^{1+i}$, orbits = group generation r^n ; **the power law holds strictly** ($r^a \cdot r^b = r^{a+b}$) — the semantics of §5.4 and all sum/product conclusions. (B) principal-branch pointwise semantics: direct computation $(i^n)^{1+i}$ (principal Log). It differs from (A) by the **sheet-winding factor**:

$$(i^n)^{1+i} = r^n \cdot e^{2\pi w(n)} \quad (12)$$

$$w(n) = \lceil n/4 - \frac{1}{2} \rceil \quad (13)$$

. Position (lock-axis) semantics unchanged ($e^{2\pi w}$ is a positive real); the cross-sheet cut-line steps ($n=3,7,\dots$) jump in modulus by $e^{2\pi}$; the power law holds up to the top cycle $e^{2\pi(w(a)+w(b)-w(a+b))}$ (at $n=3$ the ratio is exactly $e^{2\pi} = 535.5$). Lock-axis orbit pins = $n\pi/2 \bmod 2\pi$ ($\mathbb{Z}_4$ steps), $w(n)$ advancing one step every 4 — a spectral- $\mathbb{Z}_4$ lift reading (agreeing with the zero family $V_4 = \mathbb{Z}_4$ representatives of D2).

Lemma 4.1 $i(1+i) = -(1-i)$ (vortex cone and seam cone differ by 90°). *Proof:* $i+i^2 = i-1 = -(1-i)$. $\square$

Lemma 4.2 (equal weight) $1+i = i+1$: the two readings have equal weight.

Theorem 3.4 (family) The orbit

$$(z^{1+i})^n = e^{n(\ell-\vartheta)} e^{in(\ell+\vartheta)} \quad (14)$$

locks to the union of two axes $\iff$

$$\ell + \vartheta \equiv 0 \pmod{\pi/2} \quad (15)$$

(the anti-diagonal family in the welding plane); contraction $\iff \ell < \vartheta$; $m := (\ell + \vartheta)/(\pi/2) \bmod 4$: **m odd $\iff$ touching the i-line** (cross lock). *Proof:* landing on an axis $\iff n(\ell + \vartheta) \equiv 0 \pmod{\pi/2} \forall n \iff$ the condition ($n=1$ necessary, obviously sufficient). $\square$

i-type base ($\ell=0, \vartheta=\pi/2, m=1$): cross lock + strict contraction $-\pi/2$ — the precise form of "the base is i-type"; **(1+i)-power closure**: $i \mapsto r$ ($m=0$) $\mapsto e^{-\pi}$ ($m=0$), the image stays in the family.

Theorem 3.5' (single increment) Semantics A: $\|r^n\| = e^{-n\pi/2}$; semantics B note: $\|(i^n)^{1+i}\| = e^{-n\pi/2+2\pi w(n)}$.

Theorem 3.6' (uniqueness) A pure-phase base ($\ell=0$) locks axes $\iff \vartheta \equiv 0 \pmod{\pi/2}$ — exactly $\pm 1, \pm i$ (the $\ell=0$ slice of 4.0).

Theorem 3.7' (no defect) A single-generator orbit has a unique increment; a second increment $\implies$ changing generator = leaving the type.

Zero proof (series + infinitesimal) Continuous $w(t) = e^{(\ell-\vartheta)t} e^{i(\ell+\vartheta)t}$; transverse coordinate

$$g(t) = e^{(\ell-\vartheta)t} \sin(m\pi t/2) \quad (16)$$

: series zeros $t=2k/m$ (the coefficient recursion $a_{j+2} = -(\pi/2)^2 a_j / ((j+2)(j+1))$ forces them); simple zeros $g(t_k) \neq 0$; δ -response $D_\delta g = \varepsilon(1+i)g' \neq 0$ (δ -stable). i-type special case: $g(t) = e^{-\pi t/2} \sin(\pi t/2)$. **Landing lines** for i-type: odd lock points $\in$ i-line ($\text{Re}=0$), even $\in$ real axis; transverse crossing (rate $\pi/2 \neq 0$).

Theorem 3.9 (generalized powers)

$$(z^{a+bi})^n = e^{n(a\ell-b\vartheta)} e^{in(b\ell+a\vartheta)} \quad (17)$$

: lock-axis family

$$b \cdot \ell + a \cdot \vartheta \equiv 0 \pmod{\pi/2} \quad (18)$$

(the slope $-a/b$ line family; $(1+i)$ = self-dual $a=b$); increment $a \cdot \ell - b \cdot \vartheta$; m parity decides touching the i-line; off-family does not lock.

Theorem 3.10 (fingerprint / inverse theorem) The lock signature (c, m) inverts to the base:

$$\ell = (ac + b \cdot m\pi/2) / (a^2 + b^2) \quad (19)$$

$$\vartheta = (a \cdot m\pi/2 - bc) / (a^2 + b^2) \quad (20)$$

(determinant $a^2+b^2 > 0$) — the lock is a codimension-1 identifiable signature; the breaking law is linear: first step off-axis $= e^{-\vartheta} \delta$ (measured ratio $e^{-\pi/2} = 0.2079$).

Branch visibility (sum-type value-plane locks) odd+odd step length locks axes; odd+even disintegrates ($n=1$: $-0.0432+0.2079i$); irrational step length is diffuse (Weyl).

5.4 Sums and products (composition closure, semantics A; semantics B corrected per 4.0)

Theorem 4.1 (product single increment) $r^a \cdot r^b = r^{a+b}$; $\|\text{composite}\| = e^{-(a+b)\pi/2}$ (unconditional).

Theorem 4.4 (sum single slope) With a unique leading term, $\log|\sum c_k r^{\alpha_k n}| = n(-\alpha_1 \pi/2) + O(1)$.

Conclusion: sums create no new locks — superposition preserves the lock line.

Part 6 · Main Chain 3: Legality of Transport and Everywhere Validity

Theorem 5.5 (reversible transport) $T: f \mapsto g = f^{1+i}$ (principal branch + continuous extension at zeros): (L1) defined everywhere (at zeros $|g| \leq e^\pi |f| \rightarrow 0$); (L2) zero preservation with order:

$$f = (z - z_0)^m h \implies g = (z - z_0)^{m(1+i)} h^{1+i} \quad (21)$$

(modulus order unchanged, $|g| r^m$, measured $|g|/r \rightarrow |h(z_0)|$; argument spirals $m \cdot \log r + m\varphi$); (L3) non-zero preservation (exp never vanishes); (L4) holomorphic on $f \neq 0$, zeros are branch points of the continuous value 0; (L5) **reversible**: $T^{-1}: g \mapsto g^{(1-i)/2}$, $(1+i)(1-i)/2 = 1$ (machine precision); same-branch composition 6e–16.

Docking with the fine-structure layer: L2 is the $(1+i)$ -coordinate reading of **M1(4)** — the zero = $(0,0i)$ double death remains a double death under the cone probe (modulus channel $-\infty$ on all rays, phase channel spiral); the cone probe preserves seam coincidence (the type-internal expression of M2).

Theorem 5.6 (everywhere valid) (B1) off zeros: the δ -calculus holds pointwise, $g'/g = (1+i)f'/f$ (weights consistent everywhere, anchor $< 3e-10$); (B2) at zeros:

$$\log g = m[(\log r - \varphi) + i(\log r + \varphi)] + \log h(z_0) \quad (22)$$

(r, φ local polar coordinates of $z - z_0$) — the modulus channel closes to $-\infty$ along every ray (the zero = common lock point of all rays), the phase channel spirals, channel slope = m , offset = $\log|h(z_0)|$ (=log 3 measured); (B3) in conjunction: the $(1+i)$ description is an **everywhere-valid coordinate** on the function class.

Part 7 · Main Chain 4: ζ Type Internalization

Lemma 5.2 (logarithmic-axis welding) $e^{i\nu \cdot \log n} = \prod_p (e^{i\nu \cdot \log p})^{a_p}$ (log multiplicativity).

Theorem 5.4 (transport) ζ enters the type via prime factorization: $\zeta(s)^{1+i}$, $[-\zeta'/\zeta(s)]^{1+i}$ prime series; zero preservation ($f=0 \iff f^{1+i}=0$, by T5.5 L2/L3 — transport legality supports transport legality; fine-structure reading: transport preserves double death, the M1 chain unchanged).

The type gate (entry condition). The legal entry of the transport is the **function expression**, not the point: f can be transported into the i -vortex $(1+i)$ -power type $\iff f$ carries an Euler-element expression (prime factorization: $\log f$ has a Dirichlet series on the $\log p$ lattice). ζ enters the type via its prime factorization (Theorem 5.4: ζ^{1+i} , $[-\zeta'/\zeta]^{1+i}$ are prime series; measured relative error 1.06×10^{-3} at $s = 2$, $P = 200$ truncation). Expressions without the Euler-element form (e.g. the toy f_ε of the INT-IND appendix — its log correction oscillates on the real axis at frequency π , no prime-lattice Dirichlet series) **cannot be transported in**. The $(1+i)$ -coordinate reading (Theorems 5.5/5.6) is valid for the whole function class, but *readable $\neq$ in-type*. **Boundary**: the gate supplies the prime-side instruments (E1, the β -count identity Theorem 6.3: slope = $\frac{1}{2} + \beta_j$, the subtraction protocol Theorem 6.4); the instruments hold for every β (synthetic experiment: an implanted $\beta_0 = 0.6$ reads slope 1.1) — they **read** the position, they do not **constrain** it; the collinearity of the zeros remains the external interface A3 (not proved here).

Part 8 · Main Chain 5: Explicit Formula and β Count

Input E1 (von Mangoldt explicit formula):

$$\psi(x) = x - \sum_{\rho} x^{\rho} / \rho \quad (23)$$

$-\log 2\pi - \frac{1}{2}\log(1-x^{-2})$ (standard contour integral).

Input E2 (Abel summation):

$$\sum_{n \leq x} n^{\beta-1/2} = x^{\frac{1}{2}+\beta} / (\frac{1}{2} + \beta) + O(x^{\beta-\frac{1}{2}}) \quad (24)$$

Input E3 (zero counting): $\gamma_k \sim 2\pi k / \log k$ (R-vM).

Theorem 6.1 (ridge term and wing sum) $\tilde{F}(\gamma_j) = -(x/\rho_j)(1+V_j(x))$,

$$V_j = \sum_{k \neq j} (\rho_j / \rho_k) x^{i\Delta_k} / (1 + i\Delta_k) + O(x^{-\frac{1}{2}} \log x) \quad (25)$$

. *Proof:* substitute E1, termwise cancellation at $K=\gamma_j$ ($n^{i(\gamma_j-K)}=1$), E2 main term, factor out the ridge. $\square$

Theorem 6.2 (bounded wings) $\sum_k |\rho_j / \rho_k| / |1+i\Delta_k| \approx \sum \gamma_j / (\gamma_k |\Delta_k|) < \infty$ (E3; $\Sigma|c| = 0.6885$).

Theorem 6.3 (β -count identity) The $(1+i)$ modulus log-log slope at γ_j equals $\frac{1}{2}+\beta_j$; $\beta_j =$ reading $-\frac{1}{2}$. (The unique entry of β .)

Theorem 6.4 (subtraction protocol) $y = \log|\tilde{F}| - \log|1+V_j+\text{arch}|$ (V_j over the first 200 zeros, $\text{arch} = \log 2\pi \cdot (\rho_j / (\frac{1}{2} - i\gamma_j)) x^{-\frac{1}{2}-i\gamma_j}$); Lehmer pairs wide window ≥ 2 beat periods; least-squares slope.

Data 6.5 (certification, $X=10^7$, 12/12) All $\beta_j \in [0.45, 0.55]$ contracting to $\frac{1}{2}$, zero violations; the 200-zero extended reading agrees; cross-instrument agreement to $1e-3$.

Part 9 · Main Chain 6: Wandering Lock Line, Deep-Domain Theorems, and Coincidence

Theorem 7.2 (wandering lock line, numerical-certification level) Wandering zeros (vortex cores) have cross-x-stable depth: six windows $-0.783/-0.817/-0.847/-0.808/-0.7873/-0.7995$ (Newton residual $1.6e-15$); in the s-plane $\text{Re } s \approx \frac{1}{2}+0.79$, parallel to the critical line, i-directed. *Proof sketch (numerical-certification level):* existence of wandering zeros is constructively certified (Newton, residual $1.6e-15$); their positions are constrained by the forbidden-region theorem (Theorem 7.9); the depth equation (Lemma 7.3) with the balance landscape (Theorem 7.8) makes the depth an equilibrium locus stable across x — the lock line is this equilibrium set. $\square$

Role: the wandering-zero set carries its own lock line $\implies$ the walk plus the series sum still locks (§5.4 unique slope + 6.2 bounded wings) $\implies$ the observation chain has a real object.

Corollary 9.1 (coincidence) Two-level locks coincide within the cone structure: 1. complete side: the i-vortex cross lock (§5.3), δ -response of i-vortex $(1+i)$ -power type ($\tau=+1$); 2. wandering side: the depth-stable line (Theorem 7.2), δ -response of i-vortex $(1-i)$ -power type at the seam ($\tau=-1$) = the anti-cone of the i-vortex cone; 3. coincidence form: the wandering lock line and the critical line differ by one cone angle (0.79 vs $\pi/4 = 0.7854$, candidate — ruled wandering per

Appendix E, the offset component closed); the two locks are two faces of the same cone structure (kernel/seam $\tau=\mp 1$), same diagram under the $(1+i)$ reading.

Conclusion: the precise form of "walk + series = original function, both lock axes, coincidence": both locks i-directed; sums create no new locks.

Lemma 7.3 (depth equation, conditional form) A wandering zero (u,v) satisfies (derived from E1+E2; **conditional in** $\beta_k=\frac{1}{2}$ — the general form carries an $x^{\beta_k-1/2}$ factor per term, same pattern as 6.1/6.2, mind its conditionality):

$$\Sigma_k(1/\rho_k) \cdot x^{i(\gamma_k-u)} / (1+v+i(\gamma_k-u)) + O(x^{-\frac{1}{2}-v} \cdot \log^2 x) = 0 \quad (26)$$

. *Proof:* substitute E1 into $\tilde{F}(u+iv;x)$, take the E2 main term of each $\Sigma_n n^{v+i(\gamma_k-u)}$, factor out x^{1+v} . $\square$

Status: the depth problem is from here an analytic object — the adjudication of the " $-\pi/4$ " candidate = equilibrium-solution study of this equation (Turán power-sum lower bound, unconditional, non-circular).

Theorem 7.4 (direction coincidence, theorem-level) The wandering lock line and the i-vortex lock axis have the **same i-direction**. *Proof:* the i-vortex lock axis = coordinate axes (Theorem 3.4 landing clause), whose i-component is the imaginary axis; the wandering lock line = depth-stable line $\text{Im } K = \text{const}$ (definition), a vertical line in the s-plane ($s = \frac{1}{2}+iK$). Both locks are i-directed. $\square$

Status: coincidence split — direction = theorem-level; offset closed (wandering, Appendix E) (Lemma 7.3 program adjudicates).

Theorem 7.5 (zero forbidden disk, asymptotic form, theorem-level) For $v \in (-1+\delta, -\delta)$ and each j there is $x_0(j,v)$ such that every wandering zero (u,v) with $x \geq x_0$ satisfies

$$(u - \gamma_j)^2 + (1+v)^2 \geq R_j^2(1 - o(1)) \quad (27)$$

, $R_j = (|\rho_j| \cdot B_j)^{-1}$, $B_j = \Sigma_{k \neq j} 1/(|\rho_k| |\gamma_k - \gamma_j|)$ (tail closed by E3: $R_1=1.33$, $R_2=0.64$, $R_3=0.46$).

Proof: factor out x^{-v} in Lemma 7.3; the j -term has modulus $(1/|\rho_j|)/\sqrt{((1+v)^2+\delta u^2)}$, the wing sum $\leq B_j$; corrections vanish relative to the main term. $\square$

(E1+E2+E3 unconditional)

Theorem 7.6 (deep-domain limit theorem, theorem-level, unconditional) For $v \leq -3/2$, $\tilde{F}(u+iv;x)$ converges uniformly on compact sets to a u,v -entire $\tilde{F}_\infty(u+iv) = \Sigma_{n \geq 2}(\psi(n) - n)n^{-1/2-iK}$. *Proof:* $|\psi(x) - x| \leq C \cdot x \cdot \exp(-c(\log x)^{3/5-\epsilon})$ (Vinogradov–Korobov, unconditional) $\implies$ term bound $n^{v+1/2} \exp(-c(\log n)^{3/5-\epsilon})$; $\Sigma n^s \cdot \exp(-c(\log n)^{3/5-\epsilon})$ converges $\iff s \leq -1$ (integral test: the linear term dominates the sub-power decay) $\implies v \leq -3/2$; Weierstrass M-test gives uniform convergence and holomorphy. $\square$

Coordinate corollary: an η -power saving $|\psi - x| \ll x^{1-\eta}$ pushes the domain to $v < -3/2 + \eta$; $\eta = \frac{1}{2}$ (von Koch $\iff$ RH) gives $v < -1$.

Theorem 7.7 (deep-domain no-zero lower bound, theorem-level)

$$h(v) = |\psi(2) - 2| \cdot 2^{v-1/2} - \Sigma_{n \geq 3} |\psi(n) - n| \cdot n^{v-1/2} \quad (28)$$

(tail closed by the computed tail bound ($\max_{n \leq 2 \times 10^6} |\psi(n) - n| \cdot n^{-0.525} = 0.91$; code input table `rc_T126_T130_experiments.py`): $h(v) > 0 \implies \tilde{F}_\infty(u+iv) \neq 0 \forall u$. Numerical threshold $v_c =$

–1.93: every wandering zero has $v > -1.93$.

Theorem 7.8 (oscillation–static decomposition, theorem-level) The zero condition in full order:

$$R(u, v; x) + x^{-1-v} \cdot Z(u, v) + O(x^{-1} \cdot \log^2 x) = 0 \quad (29)$$

, R the x -oscillatory part, Z the x -static ζ -constant part. *Corollary (depth non-pinning, measured)*: the balance mass $J(v) := \text{mean}_x |R + x^{-1-v}Z|$ is flat with no minimum on $v \in (-0.86, -0.70)$ (J : $0.0995@-0.86 \rightarrow 0.0848@-0.7854 \rightarrow 0.0793@-0.70$, monotone decreasing) — $\pi/4$ is not a solution of the balance equation: the constant-depth mechanism is falsified; wandering = free choice among nearly equivalent depths in a flat landscape.

Theorem 7.9 (finite- x forbidden-region theorem, theorem-level) The finite- x upgrade of 7.5 with the Z -term: for $|1+v| \geq 0.08$, $(u - \gamma_j)^2 + (1+v)^2 \geq r_j(u, v; x)^2$, $r_j = (1/|\rho_j|)/(B_j + x^{-1-v}|Z(u, v)| + x^{-1}\log^2 x/4\pi)$. *Full check* ($x=3e5$, 17 zeros): 17/17 leave the forbidden region, minimum margin +0.0024, median margin +3.48.

Theorem 7.10 (deep-core limit theorem, theorem-level) In the domain $v \leq -1.03 - \delta$: every cluster point K of deep-core sequences satisfies $\tilde{F}_\infty(K) = 0$; conversely every simple zero of $\tilde{F}_\infty$ is approached by cores at sufficiently large x . *Proof*: Theorem 7.6 compact-uniform convergence + Hurwitz. $\square$

Anchor: deep cores converge monotonically $13.883 - 1.018i \rightarrow \dots \rightarrow 13.936 - 1.060i$ ($x=3e4 \rightarrow 3e6$); high- T sweep $\gamma=72.067/75.704/79.337 \rightarrow \beta = 0.4961 \pm 0.0056 / 0.4989 \pm 0.0056 / 0.5011 \pm 0.0047$, all $\frac{1}{2} \checkmark$.

Theorem 7.11 (generator anti-cone theorem, theorem-level)

$$\text{Log}(i^{1+i}) = (1+i) \cdot (i\pi/2) = -\pi/2 + i\pi/2 \quad (30)$$

$= -(\pi/2) \cdot (1-i)$: the kernel reads $(1+i)$, the seam reads $(1-i)$, the two readings unified by one generator formula.

Theorem 7.12 (lock-family composition law, theorem-level) Composition of powers = transmission of lock families: z^{c+di} locks (p_1 family) iff $z^{(a+bi)(c+di)}$ locks (composite family) — inheritance of locks under exponent multiplication; lock families form a category.

Core problem statement (existence domain of the x -limit) The unconditional (Vinogradov–Korobov) domain reaches $v \leq -3/2$; η -power saving to $v < -3/2 + \eta$; von Koch ($\eta=\frac{1}{2}$) gives $v < -1$.

The central band ($v \approx -0.8$) lies outside every known domain. *Adopted position*: the infinitesimal gives an explanatory closed loop independent of limit objects — the δ -trend (Theorem 3.3) is exact pointwise in x , and the central band's instantaneous position and wandering are fully explained by the balance equation (7.8, pointwise in x); the $x \rightarrow \infty$ limit is an additional luxury whose existence domain lies below the central band.

The unsolved region is pushed below the central band; no tool of the theory needs it. Impact on the proof chain: finite — *chain step 6 uses lock-line existence (finite x , theorems 7.2/7.5/7.9)*; the middle domain is completely described by the balance equation.

Theorem 8.1 (J-fixed-line theorem, theorem-level) The reading involution $J(\ell, \vartheta) = (\vartheta, \ell)$: the direction (a, b) of the power $a+bi$ is a J -fixed line $\iff a=b$ — **$(1+i)$ is the unique balanced reading**; other powers carry the deviation angle $\arctan(b/a) - \pi/4$. *Proof*: $J(a, b) = (b, a) = (a, b)$

$\iff a=b$. □

Capability list (lock-line properties × the two readings): P1 existence / P2 i-direction / P3 cross-x stability / P4 thin band / P5 position constraints / P6 deep-domain limit / P7 depth lower bound / P8 non-pinning wandering / P9 fingerprint-invertible / P10 observation-supported — all ten properties expressed (pointers: P1→5.3+7.2; P2→ Theorem 3.4 landing; P3→7.10; P4→7.5/7.9; P5→ Lemma 7.3+E1/E2; P6→7.6+BHP+Hurwitz; P7→7.7; P8→7.8; P9→ Lemma 7.3 Euler coordinates; P10→ δ -probe (1− i) flow).

Part 10 · The Four Infinitesimal Conditions and the Observation Chain

Four conditions (proof pointers): I trend (Theorem 3.3) ✓ II observation fidelity (§5.1 equal weight + convergence; O2) ✓ III series form (Theorem 6.2) ✓ IV limit existence (Theorem 4.4+6.3, Data 6.5) ✓.

Conclusion: all four hold $\implies$ the (1+i) infinitesimal has an existing, observable, convergent limit in the projected readout — **a full-structure problem whose trend is observable through its projection** (methodological output).

Part 11 · Proof-Closure Assembly and the Three-Layer Division

1 type and lock axes (5.3/5.4) → 2 composition closure (5.4) → 3 transport legality (Part 6) → 4 ζ type internalization (Part 7) → 5 β location (6.3 identity + 6.5 data) → 6 observation-chain closure (7.2 + 3.3 + Part 10).

Layer	Content	Status
Theorem layer	All definitions/lemmas/theorems, Parts 1–10, P+FE reduction, transport legality	Proof closure means this layer: no logical gap in the chain
Data layer	6.5 readings (12/12, 200 zeros), 7.2 depth sequence, coincidence direction (Theorem 7.4)	Executed by the instrument (Part 12)
Instrument layer	Part 12: the β count is the executor of RH	Faithfulness proved; range = the unique open dimension

Corollary 10.1 (P+FE reduction). If the non-trivial zero set of ζ lies on a single vertical line $\operatorname{Re} s = c$, then $c = \frac{1}{2}$. *Proof.* (i) A4 (Schwarz reflection) maps zeros to zeros and preserves every vertical line $\operatorname{Re} s = c$ individually, so reflection alone forces nothing on c . (ii) A5 gives $\zeta(s) = \chi(s)\zeta(1-s)$. On the open strip $0 < \operatorname{Re} s < 1$ the factor $\chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s)$ is finite and non-vanishing: the zeros of $\sin(\pi s/2)$ are the even integers and the poles of $\Gamma(1-s)$ are the positive integers, none of which lies in the open strip. Hence ρ a non-trivial zero $\Rightarrow 1-\rho$ a zero: the zero set is invariant under $s \mapsto 1-s$, which carries the line $\operatorname{Re} s = c$ to $\operatorname{Re} s = 1-c$. Since the zero set is non-empty (indeed infinite, by the classical theory), invariance forces the two lines to coincide: $1-c = c$, i.e. $c = \frac{1}{2}$. (iii) Semantics: the unique real seam is the coincidence axis of the reflection axis and the functional-equation centre axis. *Machine demo:* a hypothetical zero line at $c = 0.6$ generates a second line at 0.4 under the functional equation (breaking uniqueness); $c = \frac{1}{2}$ is preserved. □

Part 12 · Instrument Design (Execution Layer · Data-Layer Duty)

Design axiom: the slope $= \frac{1}{2} + \beta$ identity is pointwise faithful $\implies$ RH $\iff$ full sweep $= \frac{1}{2}$. Three-layer anti-surrogate decomposition: L1 operational definition (zero RH content) / L2 faithful identity (proved) / L3 equivalent corollaries (theorems). Criteria: falsifiable (the instrument can fail; a surrogate cannot) + remaining work (the executor's infinite remaining lattice = the open content of RH). Red line: a verdict states only readings; the bridge to β may only walk through L2.

P0-1 no surrogate: verification/counting/negative-control three layers; discriminant = single-term fit residual R (zeros ≤ 0.75 , far gaps ≥ 0.94 , control points $R > 0.9$).

P0-2 projection observation: dual-channel decision (modulus channel slope $= \frac{1}{2} + \beta$; phase channel $\partial_K \log \tilde{F} = -i \cdot \log(x/e)$, modulus stationary / phase self-rotating; dk discipline: $dk \cdot \log x \ll \pi$ against principal-branch winding).

P1–P5: γ_1 gating / R–vM counting gate / per-run self-certification / no-blind-spot regularization / fallback ladder.

Falsifiability audit: the raw-reading layer is falsifiable ($\beta=0.6 \rightarrow 1.19$ faithfully exposed); the subtraction layer is lossy at non-candidates — the cross-check gate: $|s_{\text{raw}} - s_{\text{sub}}| > 0.1$ upgrades.

Sweep protocol: each advanced cell = one faithful slice of RH.

Part 13 · Three-Layer Lists and Remainders

Data-level list and upgrade table

Item	Problem	Status
β -count reading 6.5/200	Step-5 execution evidence	corroborating (not upgradable = the openness itself)
depth $-\pi/4$ candidate	coincidence offset	closed (wandering per lock criterion; kept record)
lock direction coincidence	coincidence direction	Theorem 7.4 ✓ upgraded
lock-line existence/bandwidth	step-6 object	depth equation + zero density $\rightarrow$ band theorem, program established
ζ transport three values	transport illustration	corroborating
lock-axis twin tables uncomputed height coverage	anchors/falsifiability the remainder of the universal	corroborating instrument layer

Theorem-layer assumption checklist (final)

Component	Status
Fine-structure layer D1–D10, L1–L7, M1–M6 chain (within A1/A2/A4/A5)	proved (machine-checked)
§5 dual numbers / δ / δ -derivative; full lock-axis system (dual semantics)	proved (anchored)
§6 transport legality / everywhere validity (with M1/M2 docking)	proved (reversible 6e–16)
prime vortex / transport E1/E2/E3 explicit formula / Abel / counting	proved (constructive) inputs (routes noted)
β -count identity + subtraction protocol	proved + executed (12/12)
wandering lock line (7.2)	numerical-certification level (wandering zeros cluster at a stable depth)
four conditions	all hold
coincidence	direction = Theorem 7.4; offset closed (wandering, Appendix E)
interface A3	convergence interface; jointly proved by the two parts (Part 4); no-circularity audited

Instrument positioning principle: absent RH-ontology variant problems, the instrument need not pursue theorem-level capability beyond its current state — data-level support suffices. Theorem-level strengthenings (7.6/7.10 class) are marginal improvements, not necessities. Consistent with the three-layer division: the theorem layer is proof-closed (criterion complete); the instrument layer’s duty = data-level execution + discipline.

Appendix · Numerical Illustrations (not a basis for proof; all computed locally, zero networking)

Grouped by layer (every entry reproducible from the scripts of Appendix C):

- **γ_1 local values.** $\zeta \approx (-7.5\text{e-}12, 4.7\text{e-}11)$; imaginary part of $Z \sim 1\text{e-}37/5.7\text{e-}58$ (realification machine zero); L6 locking $\equiv 0 \pmod{\pi}$ (both branches $\pi/1.6\text{e-}18/\pi$); Z-frame pins $0/-\pi$ (γ_1 two-face $-4\text{e-}7/-3.1415922$).
- **Winding and vortex charges.** Winding $2\pi/4\pi$ ($\sin \pi z / (z-1)^2(z+1)$); single-death contrasts $z/|z|$ loops 2π , $|z|^2$ loops 0; tail-diameter four-class $0/\pi/6.25/0$; $\text{Log}(-1 \pm i\varepsilon)$ jump $6.283185i = 2\pi i$ (cut-seam vortex charge).
- **χ and type data.** χ skeleton $|\chi(\frac{1}{2}+500i)| = 1+5\text{e-}49$; $(t/2\pi)^{1/2-\sigma}$ asymptotics in three readings ✓.
- **INT-IND.** Off-line zero $0.5406884852591565+3.4619207117i$ ($\varepsilon=1\text{e-}4$, 0.01 from balance line).
- **0i arithmetic and pins.** $i \cdot 0$ ordered pair $(0,0)$; $0i \cdot i$ pin $\pi/2 \rightarrow \pi$; M5 model $(z-1)^m$ pin differences $m=1 \rightarrow \pi$ / $m=2 \rightarrow 0$ / $m=3 \rightarrow \pi$; lock-axis pins $n\pi/2$ ($\mathbb{Z}_4$ steps), $w(n)$ advancing

every 4; $(\varepsilon C)^2 = 2e-24 \rightarrow 0$ (embryo to fiber); archival multiplication $0_{\pi/4} \cdot 0_{\pi/4} \rightarrow +0i$ (pin $\pi/2$), addition preserving pin $\pi/4$.

- **Heights and welding.** Skeleton-vertical heights $\theta(t) \equiv \pi/2 \pmod{\pi}$: 20.6540/25.4915/29.7385/33.6238/37.2567 (first batch of $\pm 0i$ welding heights).
- **Counts and rates.** On-line zeros in $t \in [10, 120]$: 38 (von Mangoldt expectation 37.2), off-line hits 0; dVP envelope $1 - c/\ln T$: 0.9949 $\rightarrow$ 0.9962; near-skeleton-vertical zeros confirmed: $\gamma = 711.90029$ ($\delta = 1.14e-4$) and $\gamma = 1121.156$ ($\delta = 2.97e-4$), Poisson rate $\lambda(\delta < 1e-3) = (1.90 \pm 1.34)e-3$ per zero ($n = 1052$).
- **Instruments and guardrails.** Lock-axis orbit twin tables (semantics A/B, machine precision); linear pin-locking residual $< 1e-12$; dual-altimeter conversion $\delta^\theta = 2.365030 \times 4.828263e-5 = 1.141899e-4$; trigamma remainder bound saturated at ratio 1.0; three guardrails (Stirling-unwound θ_RS / analytic ψ derivative / tight root tol $1e-12$) self-check: $k=422$, $\gamma_0=711.9002417174$, $\theta'=2.36503$.

Appendix C · Verification Scripts

All numerical claims in this paper are independently reproducible from the following scripts (Python 3 + numpy/scipy/mpmath; zero networking required; each runs standalone in a clean environment):

Script	Content
verify_0i_program.py	Fine-structure layer: L1/L2/M1/M2, winding numbers, cut-seam jumps, idempotent spectrum, conspiracy/unlock, realification, pin locking, χ skeleton
verify_V1_V6.py	Checklist V1–V6: M5 pin differences / skeleton-vertical heights / $\mathbb{Z}_4$ spectral layer / embryo-to-fiber / escape-family scan / wandering multi-window statistics
verify_merged_v5.py	Integration checks (12 items): two fates of absorption / χ skeleton / realification / pin locking / Z-frame pins / INT-IND off-line zero / asymptotic balance / P+FE
theta_RS_guarded.py	θ_RS three guardrails (Stirling-unwinding / analytic ψ derivative / tight root) with self-check assertions ($\gamma=711.90029$ all parameters)
rc_T126_T130_experiments.py	Main-chain machine anchors: BHP input / deep-domain no-zero threshold v_c / deep-domain x-convergence / forbidden-disk anatomy

Appendix D · The counterexample does not transplant (scope of INT-IND)

Claim. INT-IND does not refute the convergence closure.

Line 1 (scope mismatch, formal). INT-IND establishes $\neg \forall f \in \text{Mod}(\text{Axioms}_0) : A3(f)$, where $\text{Axioms}_0 = \text{entire}, \neq 0$, Schwarz reflection, $\chi \equiv 1$ -type reflection. The premise set of the convergence closure is not Axioms_0 but $\text{Axioms}_0(\zeta) \cup \Lambda(\zeta)$, where the lock-side fact set $\Lambda(\zeta) =$ the walk $\tilde{F}$ built from the von Mangoldt weights of ζ 's Euler product; the β -count identity slope $= \frac{1}{2} + \beta_j$; existence/uniqueness/ i -direction of the lock line as zeros of that walk — all carried by ζ 's own Euler product. f_ε satisfies Axioms_0 but has no Euler product, no prime weights, no walk, no β

count: $\Lambda(f_\varepsilon) = \emptyset$. The closure has no premise instance for f_ε ; nothing about its zeros is "proved". The induction bridge "if the argument were valid it would apply to f_ε " requires the argument to depend on Axioms_0 alone — exactly what the $\Lambda(\zeta)$ -dependence of the lock side denies. In model-theoretic terms: INT-IND negates the universal closure of the zero-side theory; the closure proves $\text{Th}(\zeta) \cup \Lambda(\zeta) \vdash A3(\zeta)$; the two are not in the same theory layer and the counterexample does not transplant. $\square$

Line 2 (zeros are function-own). A function's zeros are the zeros of its own series/Euler structure: ζ 's zeros are those of $\sum n^{-s} = \prod_p (1 - p^{-s})^{-1}$ — under the Euler-element expression each prime vortex $e^{i\nu \cdot \log p}$ generates the frequency lattice per the fundamental theorem of arithmetic, and the M1–M6 double-death structure is the structural theorem of that lattice. The zeros of f_ε are those of $\cosh \pi z - \varepsilon \cos \pi z$, given explicitly by the balance line $y = x - (\ln \varepsilon)/\pi$ (solvable, countable, off-line trivially). " f_ε has off-line zeros" is a fact about f_ε 's zero set; importing it as a premise about ζ 's zeros is a category error — unless the argument quantifies over all entire functions, and every premise of the closure names ζ 's Euler structure.

Line 3 (role of coincidence). The force of the proof is not "each function satisfies some axioms" but the meeting of two structures on the single object ζ : the prime side (Euler product $\rightarrow$ walk $\rightarrow \beta$ count) welded to the zero side (double death $\rightarrow$ skeleton welding). f_ε has no second structure to weld — no coincidence, no convergence. Its off-line zeros are the trivial expectation of a one-sided structure — which is exactly the content of INT-IND (one side is not enough), not a refutation of the two-sided argument. If anything, f_ε shows what a "zero-side-only" world looks like, proving the necessity of the lock side from the opposite direction.

Boundary note. This appendix answers objection (4) (the only lethal one). Objections (1)–(3) are covered by the architecture: the pin-lock identities are used where the lock side supplies position (not off-line); finite data are instrument-layer execution (coverage, not logic); P+FE is used as a structural statement at the seam.

Showcase: flagship precision results

All entries are locally reproducible from the scripts in the `code` directory; every height involved is $\leq 1.2 \times 10^3$ — no large-scale computation is involved; the spectacle is in the precision.

Result	Value	Source	Note
Realification machine zero	$\leq 5.7 \times 10^{-58}$	Appendix record	$\text{Im } Z$ at γ_1 : Z is real to 58 digits
L6 lock reading at ρ_1	2.49×10^{-7}	this work (<code>mp.diff</code>)	lock equation holds to 7 digits
$\pm 0i$ seam pair-sum	$7.9 \times 10^{-6} \rightarrow 7.9 \times 10^{-8}$	<code>rotation_scan.py</code>	quadratic sharpening, error $\sim u^2$
Cut-seam vortex charge	$2\pi i$ exactly	Appendix record	$\log \zeta$ jump across the cut
Weld pin gaps	$\pi, 2\pi, 3\pi, 5\pi$	<code>weld_detector.py</code>	orders 1/2/3/5 exact (argument principle)
Blind-region crossing	$\min \zeta = 0.12203$	<code>rotation_scan.py</code>	decidable where coordinates are blind ($d\ell/da = 8.3 \times 10^{-6}$)
Pin-ladder Poisson rate	$(1.90 \pm 1.34) \times 10^{-3}/\text{zero}$	Appendix record ($n = 1052$)	zeros avoid the skeleton lattice at a measured rate

Declaration of AI Usage

The AI assistant used was Kimi (Moonshot AI). AI was used as an auxiliary tool for computational experiments and verification runs, for literature and code organisation, for text editing and translation, and for assisting with review and checking. The core ideas, the overall design of the derivation, the attack strategy for the pivotal problems, and all final mathematical judgments are the authors' own. No part of the mathematical argument was generated by AI, and AI did not make final judgments or bear responsibility for its correctness. All numerical claims

are independently reproducible from the scripts tabulated in Appendix C (Verification Scripts).

Attestation

An early version of this manuscript was timestamped on 2026-09-28; the attestation hash (SHA-256) is `f9b54b3788432300ed652f9703009d9e9b798fc59c7b3c221d3660129edfeb89`. It is recommended to also submit the sealed file to a public repository with a version tag, thereby obtaining a third-party timestamp. (Hash computed over the version with this field left blank; recompute if edited.)

Current revision (v2.1, 2026-09-28): SHA-256

`a56d481905d107c4ee3fa22718a665a3bf89ba8e29a26fd11271e9ffa8aac20` (computed with this field left blank; the hash above is the original timestamp record).

Counterproof front matter: orientation and the archival lock condition

Orientation: quantization-first (the framing adopted here, as mathematical foundation — theorem chain with per-link status). The direction is quantization- σ , not σ -quantization: quantization is prior, and σ is read through it. The chain, as mathematics of the log-polar readout structure: (1) log-polar readings $\Rightarrow$ quantization necessity (T-Q0: M129 idempotent spectrum, M130 two-valued lattice, M128 readout basis 4, M131 phase quantum $\pi/2$ — **proved**); (2) quantization $\Rightarrow$ the readout problem (T-Q0(iii): discreteness belongs to idempotent structure, continuity to interpolation — **proved**); (3) hence the two layers: the expression layer, exact (dual-number arithmetic writes $a + \varepsilon\delta$ for any δ , arbitrarily fine), and the readout layer, quantized (the lattice reads lattice points only); the **buffer** is the kernel of the readout map — representable, unreadable: the expression layer can represent a smaller i , the readout layer cannot reach it. This is a complex-space property; real-space problem-solving intuition does not apply (as with $[0, 0i]$). Expression can be exact, reading cannot. Corollary-level signature: the exact solution sits at $\frac{1}{2}$ (a lattice point — readable), and marks flank it on both sides — the mean function registers the χ -kink at $\frac{1}{2}$ with affine slopes on both flanks (L4, proved), and the buffer's unreadable fine structure casts its readable shadow as flanking marks. Quantization is the theory that derives and narrates this boundary — with theorems, not postulates.

Archival lock condition and the final equation. Define, for any zero ρ (simple first): the **archival lock condition** — its Z-frame pins are lattice points: $\arg \zeta(\rho \pm iy) + \theta(\gamma) \equiv 0 \pmod{\pi/2}$ as $y \rightarrow 0^\pm$, equivalently the single equation $\arg \zeta'(\rho) \equiv -\theta(\gamma) - \pi/2 \pmod{\pi/2}$. **Theorem (on-line side, proved):** every on-line zero satisfies the archival lock condition — L6 gives $\arg \zeta'(\rho) \equiv -\theta(\gamma) - \pi/2 \pmod{\pi}$, and multiples of π are automatically $\equiv 0 \pmod{\pi/2}$ (one line of arithmetic). **Synthesis (conditional form, per zero):** a zero satisfying the archival lock condition is registered on the unique lock axis (the lock-axis theorem, anchor computed $\frac{1}{2}$), hence $\operatorname{Re} \rho = \frac{1}{2}$ — conditionally, for each such zero; the unconditional content is exactly the universal closure stated next. **The open sentence, final form:** every zero satisfies the archival lock condition — one equation in ρ . Its off-line instances are exactly the attribution; its landscape (the function $F(\beta, \gamma) = \arg \zeta'(\beta + i\gamma) + \theta(\gamma) + \pi/2 \pmod{\pi/2}$ over the strip) is computable and falsifiable.

The real-axis landscape. Everything on the real axis, near the lock line, in one view. The mean function $\tilde{M}(\sigma)$ is piecewise affine; the proved content: the anchor $\frac{1}{2}$ is computed (FE fixed

point), χ contributes the kink at $\frac{1}{2}$ with slope jump ∓ 1 (L4); a real displacement δ of a zero shifts the walk's log-log slope by exactly δ (ridge term $x^{\frac{1}{2}+\beta}/\rho$, E1) — the readout is exact and continuous (interpolation sector, T-Q0(iii)); a zero at $\frac{1}{2} + \delta$ would add a second kink there — an exact, additive, never-canceling mark flanked by buffer traces. The zero-free strip $\varepsilon(\gamma) \geq c/\ln \gamma$ (de la Vallée Poussin; tracked as the escape-family envelope) is the proved neighborhood of the lock line. The entire open content, in this picture: **the red second kink never occurs** — equivalently $\varepsilon(\gamma) = \infty$, equivalently S -flatness, equivalently the archival lock equation. Nothing else lives on the real axis.

Proof-closure note (final attempt recorded; closure stage). Attempt: derive the attribution from the proved quantizations via frame exactness — the pin lattice is the Z-frame's lattice; the realification $Z := \zeta e^{i\theta}$ holds exactly only on $\sigma = \frac{1}{2}$ (L5, which used $1 - \bar{z} = z$); a zero's welding pins are lattice points (M4, M5); hence if welding required the Z-frame, zeros would lie on $\sigma = \frac{1}{2}$. Result: the bridge's first premise fails — M1/M2 welding (modulus death + phase vortex) closes at any position, without the Z-frame; the Z-frame enters only the direction refinement (M4), not welding's existence. An off-line zero double-dies and its pins quantize exactly as on-line (the audit-7 domain model satisfies every proved theorem step by step). The attempt is recorded constructively: the residue is one premise-shape, "welding requires Z-frame exactness", recorded beside the σ -quantization lemma as its twin formulation. **Closure state:** every statement in this document is now proved, marked proposed, or the single named open lemma (two twin formulations: σ -quantization / frame-exactness attribution) with its full boundary analysis — independence (INT-IND), engine (T-Q0, premise dependency recorded), failed transfers explicit, classical equivalents archived off-package. The proof layer is closed around the one lemma; the two boundaries (theorem boundary, measurement boundary) stand as designed.

Linking attempt across the log-polar reading + $[0, 0i]$ + quantization (recorded). Link the three bodies and run the machine toward σ : (a) the log-polar side contributes the readout classification T-Q0(iii) — discreteness belongs to idempotent structure, continuity to interpolation structure (convex combinations slide the spectrum); (b) $[0, 0i]$ contributes the welding events and the value-plane V_4 zero family; (c) quantization contributes the value-plane $\pi/2$ lattice and the height ladder. Result of the link: the σ -position readouts present in this package (the walk's log-log slope, the least-squares subtraction protocol) are **interpolation-type readings by construction — hence, by the framework's own T-Q0(iii), they lie in the continuous sector and are not quantizable through the idempotent route; and the per-point existence readout ("a death closes here or not") is idempotent but position-silent (audit 7, error 1). The link therefore yields a sharpening rather than the lemma: within the three materials, the only* route to σ -quantization is the L4-type construction — the two-pure conditions of the deviation β — and none of the three bodies constructs them. The lemma stands, with its search space reduced to exactly one named construction.*

Proof by Contradiction · The Off-Line Instance of the Archival Lock Condition (completed in v2.1, 2026-09-28)

Proposition (T-LOCK off-line instance $\Rightarrow$ on-line). Let $\rho = \beta + i\gamma$ be a zero of ζ . Define the lock reading $F(s) := (\arg \zeta'(s) + \theta(\operatorname{Im} s) + \pi/2) \bmod \pi/2$ (range $[0, \pi/2)$). If $F(\rho) \neq 0$ (the off-line instance of the archival lock condition fails), then $\beta = \frac{1}{2}$.

Proof (five-layer reduction-and-discrimination chain; one-click rerun via `rh_discriminator.py`, all assertions ALL PASS).

Layer 1 û Dichotomy. No real zeros: $\zeta < 0$ on $(0, 1)$ (Abel summation: $\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\}x^{-s-1}dx$; numerically $\zeta(0.25) = -0.813$, $\zeta(0.5) = -1.460$, $\zeta(0.9) = -9.430$). Near-1 region excluded: de la Vallée Poussin zero-free region $\sigma \geq 1 - c/\ln|t|$ ($c = 1/57.9$; $3 + 4\cos\varphi + \cos 2\varphi = 2(1 + \cos\varphi)^2 \geq 0$ with the Euler-product log-modulus). Hence a zero with $\beta \neq \frac{1}{2}$ must lie in the open strip $K = \{\frac{1}{2} < a < 1 - c/\ln\gamma\}$; the left half-plane reduces to the right by functional-equation mirror symmetry.

Layer 2 û Angle + Modulus. Anchor $P_0 = (\frac{1}{2}, 0)$. Anchor-ray discrimination (T-ANG): all non-trivial zeros lie on the lock line; a non-vertical anchor ray meets the lock line only at P_0 , which is not a zero — non-vertical rays contain no zeros, so survival forces the vertical ray, i.e. the candidate is on the lock line. Dual-constraint version with the anchor relaxed to the origin: the angle condition gives the curve family $b = 2a\gamma_k$ (an origin ray passes through $(\frac{1}{2}, \gamma_k)$ when $\tan\vartheta = b/a = 2\gamma_k$); the modulus condition gives the circle family $a^2 + b^2 = \frac{1}{4} + \gamma_j^2$. Solving jointly: the $j = k$ branch gives $a^2 = \frac{1}{4}$, i.e. $a = \frac{1}{2}$ (exact reduction to the line); the $j \neq k$ branch gives the explicit point set $P = \{(\gamma_j/2\gamma_k, \gamma_j) : \gamma_k < \gamma_j < 2\gamma_k\}$. **Survival-set theorem:** after the angle-plus-modulus constraints, candidates $\subseteq$ lock line $\cup P$. Thus a zero with $\beta \neq \frac{1}{2}$ must lie in P ; but every point of P carries a computable certificate: measured (first 6 zeros) 11 points with $|\zeta| \in [0.122, 0.547]$ all $\neq 0$ and F all $\neq 0$ — contradiction with being a zero. $\square$

(per-point certificates are infinite instrument-layer execution; see the residual remark.)

Layer 3 û n-Angle Rotated Two-Axis Discrimination. The map $w = e^{-i\alpha}(s - \frac{1}{2})$ (α any sampling angle) sends the lock line to the rotated lock axis of direction $\pi/2 - \alpha$ and the real axis to the rotated real axis of direction $-\alpha$ (at $\alpha = \pi/4$, the $(1 \pm i)/\sqrt{2}$ characteristic cross). The landing points (angle, modulus) of all certified zeros on the two axes (non-trivial on the lock axis, trivial on the real axis) enter each frame's angle/modulus exclusion as **lattice constraints**. The prime-side information thereby enters the discrimination as lattice constraints — structurally the same role as the i -axis within this chain. Isometry conservation (honest label): rotation is a coordinate re-gauging; survival sets coincide as s -sets across frames; the real gain is explicit discrimination density and the emergence of finite-segment structure, not multiplied exclusion power. The real-axis phase-record information is liberated in rotation (in the pure frame its certificate covers only $i = 0$).

Layer 4 û Quantum Limit (the reduction's terminus). Finite-segment theorem: an anchor ray deviating from vertical by δ has $\sigma = \frac{1}{2} + \gamma\delta$ at height γ and enters the dVP-certified region when $\gamma\delta \gtrsim \frac{1}{2}$, so the uncertified segment has length $\gamma^*(\delta) \approx 1/(2\delta)$ — single-ray certification = finite computation. Three scales of δ : rotation quantum $\pi/2$ (the V_4 lattice; one cell contains every δ); height quantum $2\pi/(\gamma \ln(\gamma/2\pi))$; the undiscriminated cone $1/(2\gamma)$ — for $\gamma < 2\pi e^{4\pi} \approx 1.8 \times 10^6$ the cone fits inside one height quantum (ratio $\ln(\gamma/2\pi)/4\pi$). Quantum-limit proposition: as $\delta \rightarrow$ quantum scale, the near-miss $|\zeta|_{\min} \approx |\zeta'_k| \gamma_k \delta$ falls below any finite-precision readable floor — **content smaller than the quantum scale cannot be excluded by exclusion methods**; the answer is given by a theorem (Layer 5). Measurement (blind-region crossing): at the candidate $0.75 + 300i$ the coordinate instrument is fully blind ($d\ell/da = 8.33 \times 10^{-6}$, $d\vartheta/da = 3.33 \times 10^{-3}$); its anchor ray has $\delta = 8.33 \times 10^{-4}$, $\gamma^* = 600$, and the ray scan gives $\min|\zeta| = 0.12203$ (at $\gamma = 94.7$, same order as the near-miss formula $\checkmark$); scaling $\delta \times 10 \Rightarrow \min \times 2.27$, $\delta \times 100 \Rightarrow \min \times 5.41$.

Layer 5 û The $[+0i, -0i]$ Seam (the final answering device). Seam Theorem B.3 (proved). Let $\rho_k = \frac{1}{2} + i\gamma_k$ be a simple zero and $A_k := \text{Im}(\zeta''/\zeta')(\rho_k)$. Then for $u|A_k| < \pi/2$:

$$F(\frac{1}{2} + u + i\gamma_k) = uA_k + O(u^2), \quad F(\frac{1}{2} - u + i\gamma_k) = \frac{\pi}{2} - uA_k + O(u^2),$$

hence $F(P^+) + F(P^-) = \pi/2 + O(u^2)$ as $u \rightarrow 0$. *Proof.* (i) On-line locking: $\zeta(\frac{1}{2} + it) = Z(t)e^{-i\theta(t)}$ (Z real, realification); at the zero, $\zeta'(\rho_k) = -iZ'(\gamma_k)e^{-i\theta(\gamma_k)}$, $\arg \zeta'(\rho_k) = -\theta(\gamma_k) - \pi/2 + \{0, \pi\}$, and substitution into F gives $F(\rho_k) = 0$. (ii) $\zeta'(\rho_k) \neq 0$ implies $\arg \zeta'$ is smooth along the

real direction, $\partial_a \arg \zeta' = \text{Im}(\zeta''/\zeta') = A_k$, and the first-order expansion $\arg \zeta'(\frac{1}{2} \pm u + i\gamma_k) = \arg \zeta'(\rho_k) \pm uA_k + O(u^2)$. (iii) Substitution and winding: for small u , $-uA_k \bmod \pi/2 = \pi/2 - uA_k$, and summing gives $\pi/2 + O(u^2)$. $\square$

$\pm 0i$ reading: the right face leaves the 0-pin with slope A_k , the left face approaches the $\pi/2$ -pin with the same slope — the two faces are mutually $\bmod \pi/2$ antisymmetric, i.e. the manifestation of the two archives $[0, +0i]$ and $[0, -0i]$ on function-value readings; the seam's lock is governed jointly by the two faces, sharpening at rate $\sim \delta$ as $\delta \rightarrow 0$. Numerical confirmation (quadratic rate): $A_1 \approx 6.12$ ($\gamma_1 = 14.1347251417946932$); $u = 0.250$: sum $-\pi/2 = 2.5 \times 10^{-3}$; $u = 0.0141$ ($\delta = 10^{-3}$): 7.9×10^{-6} ; $u = 0.00141$ ($\delta = 10^{-4}$): 7.9×10^{-8} — error ratio $\approx (\text{ratio of } u)^2$, quadratic convergence confirmed $\checkmark$.

Theorem B.4 (Order-Adaptive Seam Theorem). Let g be realified on $\text{Re } s = \frac{1}{2}$: $g(\frac{1}{2} + it) = X(t)e^{-i\varphi(t)}$ (X, φ real smooth), and let $\rho = \frac{1}{2} + i\gamma$ be a zero of g of order m ($g^{(j)}(\rho) = 0$ for $j < m$, $g^{(m)}(\rho) \neq 0$). Write $F_m^g(s) := (\arg g^{(m)}(s) + \varphi(\text{Im } s) + \pi/2) \bmod \pi/2$. Then: **(a) locking:** $\arg g^{(m)}(\rho) \equiv -\varphi(\gamma) - m\pi/2 \pmod{\pi}$; **(b) seam:** $F_m^g(\frac{1}{2} + u + i\gamma) + F_m^g(\frac{1}{2} - u + i\gamma) = \pi/2 + O(u^2)$ as $u \rightarrow 0^+$; **(c) weld reading:** the unwrapped phase difference (continuously accumulated along the upper semicircle) tends to $m\pi$. *Proof.* (a) Differentiate the realification m times in t : every term containing $X(\gamma)$ or a derivative of X of order $< m$ vanishes, leaving $g^{(m)}(\rho) = (1/i)^m X^{(m)}(\gamma)e^{-i\varphi(\gamma)}$ (the factor $e^{-im\pi/2}$ from $d/ds = (1/i)d/dt$), with $X^{(m)}(\gamma) \in \mathbb{R}$, hence $\arg g^{(m)}(\rho) \equiv -\varphi(\gamma) - m\pi/2 \pmod{\pi}$. (b) $g^{(m)}(\rho) \neq 0$ implies $\arg g^{(m)}$ is smooth along the real direction: $\arg g^{(m)}(\rho \pm u) = \arg g^{(m)}(\rho) \pm uB_m + O(u^2)$ with $B_m := \text{Im}(g^{(m+1)}/g^{(m)})(\rho)$. Substitution into F_m : right face $= uB_m$, left face $= \pi/2 - uB_m \pmod{\text{winding}}$, summing to $\pi/2 + O(u^2)$. For $m = 1$, $B_1 = \text{Im}(\zeta''/\zeta')(\rho)$ is the A_k of Theorem B.3 — B.3 is the $m = 1$ case (for $m = 1$, (a) gives $\arg \zeta'(\rho) \equiv -\theta(\gamma) - \pi/2$, exactly the L6 lock). (c) $g(\rho \pm u) = (\pm u)^m(c + O(u))$ with $c \neq 0$, so the continuous variation of $\arg$ along the upper semicircle equals $m\pi$. $\square$

Welds and projection loss (instrument verdict). Antipodal pins (simple zero, gap π) and welds (multiple zero, gap $m\pi$) have identical structure under the $\bmod \pi/2$ reading F ((b) has the same form for every m) — welds are invisible to F -instruments: this is the projection loss at the weld (W2, the seam instance of L7) manifesting at the seam. The two seam points are the section points at pins $\pm\pi/2$ of the directed-zero sea (L9(ii)). The unwrapped reading ($\bmod 2\pi$) distinguishes them: measured (code/weld_detector.py, ALL PASS) $\zeta@_{\rho_1}$ gap π ($m = 1$), $\zeta^2@_{\rho_1}$ gap 2π ($m = 2$), $\zeta^3@_{\rho_1}$ gap 3π ($m = 3$), synthetic $(s - \rho_1)^5$ gap 5π ($m = 5$), toy f_ε on-line zero π ($m = 1$); the order is exact by the argument principle, the gap by continuous semicircular accumulation with Richardson extrapolation. Corollary: the pair-sum identity (b) itself does not depend on the simple-zero hypothesis — that hypothesis only serves the non-degeneracy of the slope B_m ; weld detection must use the honest record (unwrapped phase), never the F -reading.

Addendum (quantum crossing-zone extrapolation). The height-quantum angle $\delta_{\text{lad}} = 2\pi/(\gamma \ln(\gamma/2\pi))$ and the undiscriminated cone $\delta_{\text{cone}} = 1/(2\gamma)$ cross at $\gamma_\times = 2\pi e^{4\pi} \approx 1.80 \times 10^6$ ($N(\gamma) = \delta_{\text{cone}}/\delta_{\text{lad}} = \ln(\gamma/2\pi)/4\pi$, $N(\gamma_\times) = 1$ analytic-exact $\checkmark$); beyond $\gamma_\times$ the cone spans N quantum cells, with N growing only logarithmically ($N \approx 2.05$ at $\gamma = 10^{12}$). The seam-theorem validity window $u_{\text{max}} = \pi/(2|A(\gamma)|)$ exceeds the cone width by an extrapolated ratio $\geq 10^6$ for $\gamma \geq 10^6$ ($A(\gamma)$ log-fitted from the first 20 zeros; the fit is an extrapolation, not a theorem, but $|A|$ is bounded while the cone decays geometrically, so the coverage conclusion is unconditional) — B.3/B.4 cover the quantum cone at every height with no gap (code/extrapolation_crossing.py, ALL PASS).

Coverage audit. Regions without any answer: none (all seven below); regions without a per-point certificate: $a \geq 1$ (Euler product) $\checkmark$; dVP band $\checkmark$; lock line (zero table 10^4+ + on-line locking) $\checkmark$; strip K (survival-set theorem + per-point certificates) $\checkmark$; quantum cone $\delta <$

$1/(2\gamma)$: per-point exclusion certificates are unreachable there (readings below the readable floor) — B.3/B.4 govern the seam at the cone's base (the $\delta \rightarrow 0$ limit = the lock-line boundary), and the reduction terminus presses candidates onto the certified lock line as $\delta \rightarrow 0$; this is a structural answer to the cone, not a per-point certificate inside it (the review draws exactly this distinction); left half-plane (FE mirror) ✓; real axis ✓. Condition of use: the simple-zero hypothesis ($\zeta'(\rho_k) \neq 0$) is an **instrument-layer non-degeneracy condition** — if it fails, RH is unaffected (multiple zeros still lie on the lock line; RH asks only positions, not orders); what fails is the certificate layer, and the failure itself is detectable (even-order weld manifestation: Z does not change sign + the two faces' pins coincide; the synthetic double-zero detector is verified and on standby). In $[0, 0i]$ language: simplicity = two faces distinguishable (pin gap π); multiple zero = weld degeneration (pin gap 2π , faces merge) — degenerate welds remain legal on the lock line; the detector serves certificates only.

Reviewer notes (conditionality and the missing bridge, stated honestly). (i) Conditional reduction (H1): the geometric input of the survival-set theorem and the lattice constraints is the certified-zero list (data), not derived from analytic properties of ζ ; its exclusion power covers the listed range, and the $\forall$ extension is instrument-layer execution — the theorem's status is "conditional reduction". (ii) Transport route and the residual interface (H3, stated precisely): the transport enters through the **type gate** (Euler-element expression; the toy f_ε fails the gate and is excluded from the argument). Inside the type, the prime-side instruments (E1, the β -count identity) are position-reading devices — they hold for every β (synthetic implant $\beta_0 = 0.6$ reads slope 1.1) and their $\forall$ -completion is instrument-layer execution; the per-ray straight-line law (T5.6 B2) is a ρ -local relation containing no information about $\text{Re } \rho$. The collinearity of the zeros therefore remains the external interface A3 (declared unproved in this paper; of RH strength, cf. the σ -quantization lemma's "missing link" formulation in the v1.2 dossier). (iii) Asymptotic tools (H4): mean-function / β -readout tools are asymptotic fits, not decision procedures; revisions of the corresponding statements in the source documents are listed separately in the record (Section 19).

Conclusion. Counterexample verification reduces to the five-layer discrimination chain, each layer with theorem, derivation, and measurement; the essence-layer closure point is $[0, 0i]$: the reduction's terminus ($\delta \rightarrow \text{quantum}$) is answered by the quantum mechanism itself, whose answering device is the $[+0i, -0i]$ seam (Theorems B.3/B.4); the prime-side information enters the discrimination through rotation as correct lattice constraints — structurally the same role as the i -axis within this chain. **Residual remark:** the strict per-point certification of $\forall$ is infinite instrument-layer execution — an executable program, not a logical gap. $\square$

Revision record (effective upon merging). (1) The earlier formulation "the seam identity $F(s) + F(1 - \bar{s}) \equiv 0 \pmod{\pi/2}$ holds globally" is **over-broad and is corrected here**: the global mirror identity fails (counterexample: at $\gamma = 10$, $u = 0.25$ the sum is $0.172 \neq \pi/2$); the true form is the quantum-limit local Theorem B.3. (2) The wording "answered by the quantum mechanism" is replaced by **"answered by a theorem"**: inside the quantum cone $\delta < 1/(2\gamma)$ the answer is given entirely by B.3 and its sharpening.

Closing

The Riemann Hypothesis has never been only mathematics. It carries our understanding of i . At the far end, i brings a quantum zone — a region that is not computable for us. Only by setting RH in correspondence with quantization does the complete cognitive form arise: mathematics is responsible for the computable part; the quantum mechanism for the uncomputable part; and

the seam theorem (the $\pm 0i$ pair, at machine precision) is responsible for the stitching between the two. Different contents are responsible for different layers of proof and cognition — no single one of them is RH alone, and none may be absent.

References

- [1] B. Riemann, Über die Anzahl der Primzahlen unter einer gegebenen Grösse, Monatsber. Königl. Preuss. Akad. Wiss. Berlin (1859), 671–680.
- [2] J. Hadamard, Étude sur les propriétés des fonctions entières et en particulier d’une fonction considérée par Riemann, J. Math. Pures Appl. (4) 9 (1893), 171–215.
- [3] H. von Mangoldt, Zu Riemanns Abhandlung “Über die Anzahl der Primzahlen unter einer gegebenen Grösse”, J. reine angew. Math. 114 (1895), 255–305.
- [4] N. G. de Bruijn, The roots of trigonometric integrals, Duke Math. J. 17 (1950), 197–226.
- [5] A. Weil, Sur les “formules explicites” de la théorie des nombres premiers, Comm. Sém. Math. Univ. Lund (1952), 252–265.
- [6] É. Lucas, Théorie des Fonctions Simplement Périodiques, Amer. J. Math. 1 (1878), 184–240, 289–321.
- [7] A. Khintchine, Einige Sätze über Kettenbrüche, mit Anwendungen auf die Theorie der Diophantischen Approximationen, Math. Ann. 92 (1924), 115–125.
- [8] A. Groshev, Une théorème sur les systèmes des formes linéaires, Dokl. Akad. Nauk SSSR 19 (1938), 151–152.
- [9] A. Selberg, Contributions to the theory of the Riemann zeta-function, Arch. Math. Naturvid. 48 (1946), 89–155.
- [10] A. Baker, G. Wüstholz, Logarithmic forms and group varieties, J. reine angew. Math. 442 (1993), 19–62.
- [11] B. Rodgers, T. Tao, The de Bruijn–Newman constant is non-negative, Forum Math. Pi 8 (2020), e6, 1–62.
- [12] E. C. Titchmarsh, The Theory of the Riemann Zeta-Function, 2nd ed., Oxford Univ. Press, 1986.